Access to Opportunity in the Sciences: Evidence from the Nobel Laureates*

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December 2025

Abstract

Barriers to scientific careers could inhibit discovery and limit human progress. We study unequal opportunity by studying the childhood SES of Nobel laureates in the sciences. The average laureate grew up in an 87–90th percentile household in their country, and at the 95th percentile globally. Representation of the bottom 90% doubled from 1901–2023. Female laureates had more elite parents, and U.S.-born laureates came from more humble backgrounds than those born elsewhere. Global inequality in access to the sciences has not improved. A model explains the circumstances under which these inequalities matter for misallocation and scientific progress.

*This project was funded by Fast Grants for Innovations in STEM Talent Identification and Development. This paper benefited from excellent research assistance from Benjamin Levesque and Charlie Rudge. We are grateful for very helpful discussions with Kipp Bradford, James Feyrer, Matthew Gentzkow, Benny Goldman, Sasha Gusev, Ryan Hill, Adam Looney, Emir Kamenica, Douglas Staiger, Bruce Sacerdote, and Heidi Williams.

"I am, somehow, less interested in the weight and convolutions of Einstein's brain than in the near certainty that people of equal talent have lived and died in cotton fields and sweatshops."

— Stephen Jay Gould

1 Introduction

The question of who becomes a scientist—and whether societies make the most of their potential scientific talent—has attracted interest for over a century (Galton, 1874; Holland, 1957; Zuckerman, 1967; Merton, 1968; Berry, 1981; Morgan et al., 2022). Most quantitative work on access to scientific careers has focused on entry into the sciences or on the overall composition of researchers (Morgan et al., 2022; Stansbury and Schultz, 2023; Airoldi and Moser, 2024; Abramitzky et al., 2024; Stansbury and Rodriguez, 2024). But the social origins of those who reach the very top of the scientific hierarchy—and the frontier of discovery—are less studied.

In this paper, we propose a parsimonious quantitative measure of access to opportunity in the sciences: the average childhood socioeconomic status (SES) rank of the Nobel laureates in physiology, chemistry, physics, and economics. The laureates are scientists who reached the peaks of their respective disciplines and produced discoveries of substantial benefit to humankind. In a world where access to opportunity is equal and outlier talent is randomly assigned, the average laureate would have an SES rank of 50. By contrast, in a world where scientific careers are accessible only to children of the top 1%, regardless of their share of top talent, the average laureate childhood SES rank would be above 99. Once we think about this spectrum, it is natural to wonder where our world falls within it.

We proxy childhood socioeconomic status of the Nobel laureates with by collecting historical data on their fathers' occupations and assigning them each a 0–100 socioeconomic rank score using U.S. Census data.[1] Occupational measures like these are widely used in historical studies on upward mobility when long panel data on individuals are unavailable (Duncan, 1966; Björklund and Jäntti, 1997; Long and Ferrie, 2013; Olivetti and Paserman, 2015; Song et al., 2020).

[1] We created separate ranks based on the father's income and the father's education (both predicted from occupation), and we analyze each separately. Predicting SES from mothers' occupations is less useful, due to low female labor force participation in our study period.

Our measure lets us answer two core questions. First, what is the average childhood socioeconomic rank of the Nobel laureates? Second, how has this rank changed over the 125 years since the first Nobel Prize was awarded? The answer to this question sheds light on how access to opportunity has changed in the long run—and whether human societies and institutions have improved their ability to harness our collective talent in the last century.

We show that Nobel laureates are drawn from families in the right tail of the income distribution, and even more so from the right tail of the education distribution. The average laureate has a father at the 87th income percentile and the 90th education percentile. About 50–60% of laureates are from top 5% families. The most common father occupations of laureates, in order, are business owners, physicians, professors, and engineers.

The changes over time are substantial. The average winner was from a 92nd income percentile family in 1901, and from an 84th income percentile family in 2023; the mean education rank fell from 94 to 86 over the same period. Alternatively, the share of laureates from the bottom 90% of the income distribution rose from 23% to 37% (and from 20% to 33% for education). Access to scientific opportunity, while still unevenly distributed, is considerably more equal than at the beginning of the 20th century. There are no significant differences in levels or trends across scientific fields.

Female laureates come from more elite families (91st income percentile, compared with 87th for men), suggesting that some of the barriers to advancement in the sciences faced by women have been mitigated by family resources. U.S.-born laureates come from lower SES ranks, suggesting broader access to careers in the sciences in the United States than in the rest of the world. This is consistent with historical evidence of high intergenerational mobility among the broader U.S. population before 1940 (Long and Ferrie, 2013; Jácome et al., 2025), the era when 75% of laureates were born.

Finally, we extend our primary measure to include country of birth in addition to occupation. The results above are based strictly on father occupations; they ignore the fact that a child at the 80th percentile in the U.S. has better access to resources than an 80th percentile child in India.

We calculate global childhood SES ranks by ranking individuals in a synthetic global income distribution, which takes into account father occupational earnings and national incomes.[2] The average laureate is at the 95th global income percentile. This rank has fallen by a marginally significant 2 points over the last 125 years; the global talent pool from which the most elite scientists are drawn is small and has not broadened much at all since the 19th century.

Do these results necessarily imply misallocation and lost scientific discovery? To explore this question, we write a simple model where innate talent is heritable, and well-off parents make productive educational and unproductive signaling investments in their children.[3] We show that misallocation is most severe if innate talent has low heritability and parent investments are mostly unproductive — in this scenario, there are many "Lost Einsteins." Misallocation is least important when parent investments in children are more productive, high quality education is a good substitute for innate talent in the output process, and innate talent is both correlated with parent income and highly heritable. In this scenario, children of the rich are more likely to become scientists, but they are also the ones with the highest potential scientific productivity.[4]

The model demonstrates that the unequal distribution of laureate childhoods is not sufficient to demonstrate misallocation or unfairness in access to scientific positions. To make progress, we present a simulation in Appendix C which shows that under reasonable parameters, genetic inheritance can explain about a third of the inequality in laureate childhoods, a result notably similar to Bell et al. (2019), who find that childhood test scores (an imperfect proxy for innate ability) explain less than a third of the rich-poor gap in patent production. This result is at most suggestive, as we have wide priors on some simulation parameters, including the extent to which scientific output depends on talent vs. training, the share of parent investments that are productive, and the

[2]Specifically, we scale father occupational earnings (based on U.S. data) by a country's GDP ratio with respect to the United States. We then re-rank all individuals globally, taking into account differences in national occupation distributions.

[3]Signaling investments stand in for all parental investments that boost career progression without increasing scientific productivity.

[4]While we do not model it, welfare losses of misallocation are also smaller if scientists are close substitutes for each other. Star scientists have causal effects on scientific output (Mindruta et al., 2025; Azoulay et al., 2010), but patent/discovery races are also widespread, suggesting some substitutability (e.g. (Hill and Stein, 2025a,b)). Azoulay et al. (2019) finds that deaths of eminent scientists result in *more* creative output by competitors, more evidence that the work of some stars is substitutable.

heritability of innate talent.[5] The simulation disciplines the nature/nurture question by clarifying how assumptions about heritability and other factors translate into unequal outcome distributions.

The study of parental occupations of scientists goes back as far as Galton (1874) and has been of perennial interest. Prior work has shown that scientists' parents are disproportionately likely to have PhDs, white collar jobs, or work as educators or in the sciences (Akcigit et al., 2017; Zuckerman, 1967; Berry, 1981; Morgan et al., 2022).[6] Our study makes three primary contributions. First, we develop a parsimonious quantitative measure for summarizing and examining changes in access to scientific opportunity over time: the childhood SES rank of top scientific achievers. We focus on the most elite scientists making breakthrough discoveries, for whom misallocation could be most socially consequential (Romer, 1990; Aghion and Howitt, 1992; Galor and Tsiddon, 1997; Griffith et al., 2004; Akcigit et al., 2017).[7] Second, we develop a long time series and provide new evidence that access to scientific opportunity has expanded substantially from 1901 to the present. Third, we provide a model which clarifies the circumstances under which a high correlation of parent income and scientific output does and does not result in misallocation and welfare loss, and provide a path to exploring the role of heritability of talent.

In related work, Stansbury and Rodriguez (2024) study current relationships between career success in academia (specifically in economics in Stansbury and Schultz (2023)) and parental education, finding a substantial class gap, but do not examine changes over time. Several other papers study racial and gender gaps in academic advancement and recognition (Ginther et al., 2011, 2018; Sarsons et al., 2021; Hengel, 2022; Ross et al., 2022; Koffi et al., 2024). There is a long literature on intergenerational mobility, which has most often focused on measuring average

[5]We simulate a world where children have equal opportunities, and unequal outcomes arise only from talent differences, which are partially inherited from parents. Parent income and child scientific achievement are correlated through genetic talent. We classify Nobel laureates as the highest achieving scientists in their generation, and then compare the simulated distribution of laureate father income to the empirical distribution calculated above. Note that we run the simulation against the distribution of father occupation alone, *i.e.* ignoring the cross-country inequality; the latter inequality is self-evidently worse and is even less explained by genetic factors.

[6]There is also work studying the career paths of scientists (Holland, 1957; Hargittai, 2002; Azoulay et al., 2017). Top scientists have on average attended elite colleges and graduate programs, but these are joint signals of privilege and high early ability, which are difficult to disentangle. Focusing on the laureates' childhoods isolates a measure of privilege that they were born with.

[7]Zuckerman (1977) notably reports father occupations of the 105 U.S.-resident pre-1972 laureates, similarly finding excess representation of business owners and other high-income careers.

outcomes for children born into poor families (Solon, 1992; Clark and Bono, 2016; Chetty et al., 2014; Zimmerman, 2019; Jia and Li, 2021). Given the high social value of scientific discovery, it is independently useful to understand whether and when individuals from humble circumstances can become frontier scientists; it matters not only for equality of opportunity but could affect the direction and quantity of discovery itself (Murphy et al., 1991; Hsieh et al., 2019; Hoogendoorn et al., 2013; Freeman and Huang, 2015; Hunt and Gauthier-Loiselle, 2010; Gaulé and Piacentini, 2013; Koning et al., 2021).

Our paper is related to concurrent work by Airoldi and Moser (2024) and Abramitzky et al. (2024) on the SES backgrounds of U.S. scientists and academics from 1920–60. Airoldi and Moser (2024) study star scientists, finding that high SES predicts entry into the sciences and continues to drive professional success throughout the career cycle. Abramitzky et al. (2024) find similar results for the SES background of professors at U.S. universities, and find little progress from 1920–60. Our work focuses on the most elite scientists who have made the largest breakthroughs; we find that they are from even more elite families than the samples in Airoldi and Moser (2024) and Abramitzky et al. (2024). Relative to the prior work on scientists, to our knowledge ours is the first that measures how their socioeconomic representation has changed from the late 1800s to the present; even in the mobility literature, it is rare to find a 125-year time series with a comparable measure of success or childhood SES over time in a socially important domain.[8]

Our work also relates to a broader set of work connecting childhood and early career circumstances to scientific achievement. Bell et al. (2019) study the childhood socioeconomic ranks of U.S. inventors in a similar quantitative framework to ours, while Kim and Moser (2021) examine the role of motherhood in scientific careers. A related literature studies the networked relationships between scientists who receive major prizes (Ma and Uzzi, 2018; Jin et al., 2021).

[8]On intergenerational mobility, see Solon (1992) and Chetty et al. (2014). Studies of long-run changes in intergenerational mobility are uncommon due to data limitations, and often use creative methods like classifying SES based on individuals' names (Olivetti and Paserman, 2015; Song et al., 2020; Chetty et al., 2017).

2 Biographical Data and Measurement of Socioeconomic Rank

We collected biographical information on prize winners from 1901–2023 from a range of historical sources, covering 739 winners in Chemistry, Physics, Medicine, and Economics.[9] We exclude the prizes for Peace and for Literature, as accomplishments in these domains are more subjective and are sometimes explicitly awarded to individuals who faced challenging childhood circumstances. For each winner, we collected demographic information and parents' occupations.[10] We supplemented the data by contacting living winners directly to request information about their parents. In total, we were able to identify fathers' occupations for 711 out of the 739 laureates, but mothers for only 181; our analysis therefore focuses on father occupation as a measure of childhood socioeconomic status.[11]

Using father occupation as a measure of childhood socioeconomic status is a common approach in the historical intergenerational mobility literature (Long and Ferrie, 2013; Song et al., 2020; Long and Ferrie, 2007). Father occupation is a good predictor of other measures of socioeconomic status, and is often the only such variable available in historical sources. We use IPUMS data from the 1850–1990 U.S. population censuses to calculate occupational earnings and education levels covering the birth years of the majority of winners. For the 12 winners born before 1850, we used occupational data from 1850. Years of education and income are recorded in the population census beginning in 1940 and 1950 respectively; we use 1940 measures for all earlier periods.[12] We mapped fathers' occupations to the 269 unique occupational categories classified in the 1950 census; where there was ambiguity over occupation classification, we tested robustness to alternate coding approaches.[13]

We calculated the mean income and education ranks for working-age men in each occupation

[9]The Economics prize was established only in 1968; results are very similar if the field is excluded.

[10]Biographical information on winners was collected from biographies and interviews on the official web site of the Nobel Prize, other interviews with Nobel laureates, entries from *Encyclopedia Britannica*, profiles published by winners' academic institutions, and articles describing early life found in various other biographical sources.

[11]Mother's occupation is also less predictive of socioeconomic status than fathers, as many educated women were out of the labor force, particularly in the earliest years of our sample.

[12]The census recorded information on literacy from 1850–1930, but we elected not to use these data, given their substantial inaccuracies (see, for example, Collins and Margo (2006)).

[13]Appendix B.2 describes additional details on the calculation of occupational earnings and education levels.

category in each decade, and assigned these ranks to prize winners' birth decades and fathers' occupations.[14] These ranks reflect the average income or education rank that would have been held by an adult male in the given occupation in the birth year of the prize winner. Income and educational ranks are highly but imperfectly correlated; some example outliers are members of the clergy and (to a lesser extent) teachers (high education but low income ranks), and tradesmen (high income but low education ranks).

Some occupations have wide income distributions in the data; for instance, in 1850, farmers represented 65% of the American workforce, some of whom were very rich and others very poor. Entrepreneurs are also difficult to classify; the biographical data tells us a father is a business owner, but is often vague about the scale of the business. To the extent that prize winners are disproportionately born to the highest SES individuals within an occupation, our measure would bias our rank estimates downward, raising the observed equality of opportunity. We therefore show that our results are robust to excluding or alternately ranking these difficult-to-rank occupations.

Occupational rank data are not available for all countries in which prize winners were born, so our estimates are based on the U.S. occupational rank distribution. We adjusted for the different occupational distributions in less-developed countries by matching country-decade pairs to the U.S. decade with the most similar level of development. For example, in 1960, a professor in India would have a higher relative socioeconomic rank in India than a professor would have in the U.S.. We would therefore classify the Indian professor using the U.S. rank distribution from 1850, the year in the U.S. data with the most similar per capita GDP to India in 1960 — thus assigning them a higher socioeconomic rank within their country.[15]

We describe these as *national* or *domestic* rank measures, as they describe the relative status of winners' fathers within their own countries. These measures do not reflect cross-national differences in opportunity. To address income differences across countries, we take the following process to

[14] While the earnings and education level associated with each occupation are held constant for all years before 1940, the ranks change over time because of changes in the distribution of occupations in the country.

[15] The assumption that every country at a given income level has the same occupational distribution is of course imperfect, but it is an improvement over the alternative of using the U.S. distribution for all countries. Since there are few winners from LMICs, and high-income country occupation distributions are similar, alternate adjustments have little bearing on the findings.

rank each country-decade-occupation in the global income distribution. For each country-decade pair, we create a synthetic distribution of individuals, reflecting the occupation distribution in the U.S. decade with the nearest per capita GDP, as above. We assign each person an occupational income equal to the 1950 U.S. census income for that occupation, multiplied by the proportional GDP difference between that country and the United States, in the given decade. We then rerank all individuals in this synthetic global income distribution, creating a new global income rank for each country-decade-occupation group. For example, a tailor in 1950 has a U.S. occupational income rank of 55 — their position in the U.S. national income distribution. According to our global measure, a U.S. tailor in 1950 would be at the 97th percentile globally, reflecting high U.S. incomes compared with the rest of the world. Conversely, we would classify an Indian tailor in 1950 at the 46th global percentile, reflecting India's relative poverty in 1950. See Appendix B.3 for additional details and examples.

3 Results

3.1 Results: SES of Nobel Laureates from 1901–2023

The analysis dataset describes 711 winners from 44 countries over a period of 123 years. About 4% of winners are women, 35% are U.S.-born, and an additional 12% are from the United Kingdom.[16]

Figure 1 plots the distribution of national father income percentiles. 67% of winners have fathers from above the 90th income percentile in their birth country; 75% have fathers above the 90th education percentile. In both cases, most winners are from the top 5%. The mean laureate has a father at the 87th income percentile, and the 90th education percentile.

Appendix Figure A1 shows the ten most frequent occupations held by the fathers of laureates, along with the occupation's population share. Business owners, physicians, professors, and scientists feature most prominently. Relative to their occupational shares in the census data, professors, physicians and scientists are over-represented by a factor of 100 or more, while business owners

[16]There are six laureates whose fathers were also Nobel laureates — one of whom (William Lawrence Bragg) shared the prize with his father (William Henry Bragg) — and one winner (Irène Joliot-Curie), whose parents were both laureates.

are over-represented by a factor of 2.5. The frequency distribution of father occupations is largely similar across the different prize categories (Appendix Table A1).

We next examine how the relative childhood status of prize winners has changed over time. Panels A and B of Figure 2 shows the average income and education ranks of winners in 5-year bins since the first Nobel Prize, along with a linear estimation of the time trend. An equivalent regression estimation of the time trend is displayed in Appendix Table A2. Every 100 years, on average, there is a 6 percentage point ($p < 0.01$) decline in the average national income and education rank of laureates' fathers. The change is not driven by low-rank outliers; Appendix Figure A2 shows changes over time in the share of laureates from below the 80th and 90th SES percentiles, which have increased by 50% to 150% over the sample period. In short, recent laureates are drawn from a broader set of socioeconomic backgrounds than earlier laureates. The pool of winners is about twice as socioeconomically broad as it was in 1901. Present-day winners are still highly elite; more than 50% of 2000–2023 laureates are from top-5% families.[17]

3.2 Results: Heterogeneity by Gender and Region

3.2.1 Results: Inequality by Gender

Female winners come from more elite families than male winners; the average female laureate has a father at the 91st percentile, compared to the 87th percentile for the average male winner (Table 1, Panel A). Figure 3 compares the full distribution of father education and income ranks for both genders; the excess mass at the highest ranks for women is evident in both density functions.

By every measure (mean father rank, share of winners in the top 10% of SES, and share of winners in the top 5% SES), female laureates come from more elite families than male laureates (Table 1, Panel A). The differences are not all statistically significant, owing to the small sample of women—only 28 laureates are female.[18] Appendix Figure A3 shows the most common father

[17] Appendix Tables A3 and A4 show that these level and change estimates are robust to alternate approaches to occupational categories with more subjective rank classification, like aristocrats, farmers, business owners, and others, to excluding economics prizes, and to using (historically) observed education rather than occupational education.

[18] The distribution of women's SES ranks is more elite in education than it is in income; many female winners are in income percentiles 90–95 and education percentile 95–100. But the small sample of women suggests caution in

occupations among the set of female laureates. The main difference in occupational category is that female laureates are less likely than male laureates to have fathers who were business owners or managers, but note the small sample qualifications above.

Appendix Figure A4 plots the proportion of winners below the 90th and 95th income and education percentiles, by decade. While the graphs are consistent with the distribution of male and female ranks becoming more similar, the sample of women is too small to make a strong statistical claim about changes over time.[19]

The raw share of female winners (4%) already makes it evident that women have faced greater barriers to success in the sciences. Our results suggest that high socioeconomic status can counteract some of these barriers, but it means that elite women in the sciences are drawn from an even smaller socioeconomic pool than men.

3.2.2 Results: Inequality by Field

Table 1B shows differences in the childhood socioeconomic status of Nobel laureates by prize category. The average ranks of winners are notably similar across fields, with all fields within two rank points of another on both mean measures, and no statistically significant differences.[20]

3.2.3 Results: Inequality by World Region

Table 1C shows the regional pattern of winners' ranks. U.S.-born winners tend to come from less elite backgrounds than winners born in Europe and the rest of the world, suggesting more equal access to opportunity in the sciences in the United States, at least in the century before 1960 when most of our winners grew up. The results are consistent whether we look at family education or income ranks, and the regional differences are highly statistically significant ($p < 0.01$). Appendix Figure A5 shows the average father rank of U.S.-born vs. non-U.S.-born winners over time, showing

interpreting this finding. Appendix Tables A5 and A6 show that these estimates are robust to alternate father coding choices as above.

[19] In Appendix Table A7, we regress a rank measure on a female indicator, and an interaction between the female indicator and a post-2000 indicator. The interaction is consistently negative, but is statistically significant in only one out of six specifications, and is sensitive to alternate choices about coding specific laureates or choice of threshold year.

[20] Appendix Table A8 shows linear trends estimated separately for each field.

the consistent pattern of U.S. winners coming from less elite backgrounds.[21]

3.3 Results: Accounting for Income Differences Across Countries

So far, we have captured the dimensions of inequality that are correlated with father occupation, but have ignored the substantial differences in opportunity across countries. We now examine the global income rank estimates which take into account both the father occupation and the GDP per capita of each winner's childhood country.[22] Panel C of Figure 2 shows the average global income rank of Nobel laureates from 1901–2023, with the national income rank estimates from above as a benchmark. The results are stark. The average global rank of a prize winner is 95; this has improved only by a marginally significant (p=0.049) two rank points over the 122 years since 1901. The estimates imply that the vast majority of the world's best scientific talents are growing up in contexts that prevent them from achieving their potential and sharing it with humanity, and that little progress has been made to close the cross-country component of this gap.[23]

4 Discussion

4.1 Model: When does a lack of representativeness imply less scientific progress?

A lack of pathways to scientific careers for children born into low-income households is most concerning if it slows the pace of scientific progress. Much of the writing on inequality in the sciences focuses on Stephen Jay Gould's idea of "Lost Einsteins", people with potential to make breakthrough discoveries, but who do not receive enough education or parental inputs to break into the sciences. In this section, we present a model that clarifies the circumstances under which unequal representation identifies misallocation of talent and lost scientific output.

We consider a continuum of individuals indexed by i, with latent lifetime potential scientific

[21] Appendix Table A9 shows regression estimates from just the United States against the rest of the world. Appendix Table A10 shows the list of countries by classification region.

[22] We define the winner's country as the country where they permanently resided at age 6. See Appendix B.1 for additional details.

[23] Appendix Table A11 shows nearly identical results if we exclude the country GDP-based adjustment of the occupational rank distribution described in Section 2.

productivity Y_i. A set of gatekeepers allows only a subset of them to enter the sciences and realize that productivity. Productivity depends additively on: (i) innate talent or ability T_i; and (ii) productive investments in human capital I_i^P:

$$Y_i = \beta_T T_i + \beta_I I_i^P, \qquad \beta_T, \beta_I > 0. \tag{1}$$

Productive investments are characteristics of an individual's environment which increase their scientific productivity, and include their quality of schooling, parental investments in their health and education, transmission of cultural characteristics conducive to success, etc.[24]

We model the parent-child talent relationship using a standard heritability model. An individual's innate talent has an additive genetic component A_i, and a genetic lottery component L_i:

$$T_i = \gamma A_i + (1-\gamma) L_i, \qquad \gamma \in [0,1]. \tag{2}$$

A_i is the parent's innate productivity genotype, which is passed to the child with some mean reversion. L_i reflects the kinds of non-additive genetic deviations that make two children of the same parents differ.[25] L_i is independent of parent characteristics. Only A_i is reliably correlated with parents' talent and γ varies across traits.[26]

Families make two kinds of investments in children: a productive investment I_i^P that raises an individual's productivity and a signaling investment I_i^S that increases access to scientific careers but does not affect Y_i directly. Productive investments could be educational, such as private schooling or science camps. We use signaling to stand in for a broader set of unproductive investments that give the children of high SES parents better career opportunities. This could include nepotism through family networks, family prestige and legacy admissions, cultural capital, and any investments that

[24] Productivity is plausibly influenced by an interaction term between talent and investment (i.e. schooling increases productivity more if you have high cognitive skills). We omit it from the model for parsimony, as it would not change the nature of our conclusions below.

[25] L_i includes Mendelian segregation and recombination, interactions between genes (dominance and epistatic effects in the genetic literature), and new mutations.

[26] With the normalization $\mathrm{Var}(A_i) = \mathrm{Var}(L_i) = 1$, narrow-sense heritability from the genetics literature is $h^2 = \frac{\gamma^2}{\gamma^2 + (1-\gamma)^2}$.

improve credentials without affecting skills. Many investments have both productive and signaling components; the productive components go into I^P and the unproductive components go into I^S. Parents with high innate productivity have more resources R_i. Both kinds of parent investments are proportional to parent resources:

$$I_i^P = \alpha_P R_i, \qquad I_i^S = \alpha_S R_i, \qquad \alpha_P, \alpha_S \geq 0. \tag{3}$$

We assume parent resources are linear in parent innate talent and without loss of generality, we normalize resource units such that $R_i = A_i$.

Access decisions are made by gatekeepers who observe a career index that depends on latent scientific productivity and signaling:

$$C_i = \delta I_i^S + (1 - \delta) Y_i, \qquad \delta \in [0, 1]. \tag{4}$$

Because training positions and science funding are scarce, only the top $\theta \in (0, 1)$ share of the C_i distribution enter scientific careers. In practice, the path to an elite scientific career depends on many rounds of gatekeeping, which are multiplicative; passing any given round gives an individual additional signaling and educational resources for the next round. In later rounds of gatekeeping (e.g. major grants, scientific prizes), productivity is observable and there is substantial weight on Y_i, *i.e.* δ may be small. But in earlier rounds (e.g. elite preschool applications), gatekeeping is heavily influenced by signals and δ is high. For simplicity, we write a model with only one stage of gatekeeping. A planner with full information would select entirely on scientific productivity (i.e. setting $\delta = 0$), but gatekeepers do not observe the separate parts of the career index.[27]

Substituting these expressions gives us closed form representations of the productivity measure

[27] We ignore careers in other fields. The implicit assumption is that externalities of scientific productivity are very high.

and the career index that depend only on the primitive terms R_i and L_i:

$$Y_i = [\beta_T \gamma + \beta_I \alpha_P] R_i + \beta_T (1-\gamma) L_i \tag{5}$$

$$C_i = (1-\delta) Y_i + \delta \alpha_S R_i \tag{6}$$

Scientific productivity is correlated with parent resources through the genetic endowment ($\beta_T \gamma$) and through productive parental investments ($\beta_I \alpha_P$). It is also correlated with the genetic lottery term. The career index loads on both of these elements with the signaling factor $(1-\delta)$, and additionally loads on the uninformative signal, which is also correlated with parent resources. The social planner would assign scientific careers to the most productive individuals — those with the highest Y_i. Gatekeepers instead assign them based on C_i, so misallocation is increasing in the extent that the rank order of C_i differs from the rank order of Y_i.

The correlation between career results C_i and parent resources R_i, the main empirical object of this paper, is positive even in the social planner equilibrium. The equilibrium with gatekeepers has an additional term on R_i, making the correlation stronger.

We highlight the key intuitions of the model by focusing on the extent of misallocation in four edge cases.

Case 1: If gatekeepers perfectly observe latent productivity ($\delta = 0$), there is no misallocation. In this case, the career index collapses to the productivity index, and we get the social planner equilibrium.

Case 2: If innate talent is irrelevant ($\beta_T = 0$), there is no misallocation. In this case, scientists are made and not born — scientific capability depends entirely on educational investments and not on any inborn talent. We get the following equations for productivity and career entry:

$$Y_i = \beta_I \alpha_P R_i,$$

$$C_i = \Big[\delta \alpha_S + (1-\delta)\beta_I \alpha_P\Big] R_i.$$

Both terms depend only on R_i and thus have the same ordering. The allocation can be considered unfair, because equally talented children from poor families will not get a chance to enter the sciences, but there are no "Lost Einsteins" because the children of the rich have the largest educational investments, and innate talent is unimportant for scientific discovery. Gatekeepers (incorrectly) use the signal for decision-making. But because the signal is perfectly correlated with productive educational investments, there is no misallocation.

Case 3: Misallocation is worst if innate talent is not heritable ($\gamma = 0$):

$$Y_i = \beta_T L_i + \beta_I \alpha_P R_i,$$

$$C_i = [\delta\alpha_S + (1-\delta)(\beta_I \alpha_P)] R_i + (1-\delta)\beta_T L_i.$$

Without the correlation between parent resources and child innate talent, the career index has less in common with the productivity index, and we have more misallocation. Career success is still correlated with parent resources, but through the educational investment channel only.

Case 4: If parents invest more in signaling ($\alpha_S \uparrow$), misallocation is worse. Signaling in the model represents all parental investments which boost the career index without improving productivity. These investments increase the misallocation rate and raises the correlation between career outcomes and parent resources.

Summary. In all four cases, the career index is correlated with parent resources, but for different reasons. The extent of misallocation depends on whether this correlation arises through parental transmission of innate talent, parental transmission of educational resources, or signaling and other wasteful investments.

Unfortunately, the literature provides limited guidance on the model's key parameters. The extent to which scientific (or any other) productivity depends on talent vs. training is an age-old question. The same is true for the extent to which parent investments are pure signal (α_S), and the effectiveness of those signals for access to the sciences (δ). The literature provides many estimates

on the heritability of cognitive ability, but there is substantial dispute on which of those are most credible and what they measure.

In Appendix C, we present a simulation of an equal-opportunity world where parent income affects scientific achievement only through the genetic transmission of ability, with all investment- and access-based channels shut down. We calibrate the simulation to the relevant U.S. birth cohorts and vary key parameters (*e.g.*, heritability, assortative mating, and the mapping from ability to prizes) across mainline, nature-leaning, and nurture-leaning scenarios, and then compare the implied father SES distribution of "laureates" to the empirical distribution. Under our mainline parameterization, genetic transmission of ability can explain about one-quarter of the SES inequality in laureates' family backgrounds.[28] The aim of the simulation is to clarify that beliefs about the "natural" inequality in elite background implies assumptions about primitive parameters which can in principle be measured directly. Future work measuring those primitive parameters could help us make progress on interpreting the empirical inequality.

4.2 Other concerns about misallocation and social cost

Our evidence suggests that there is a large number of "missing scientists"—people who could have made it to the elite ranks of the sciences, but did not receive the complementary inputs required over their lives to do so, especially in lower- and middle-income countries. Here, we address several other considerations which could affect the implied misallocation and social cost of inequality in science.

First, are missing scientists producing equally valuable outcomes in non-scientific domains, such as business or medical practice? There are three reasons to think their social contributions in other fields are likely to be smaller than they would be in the sciences. First, the social benefits of research in the basic sciences are plausibly among the highest out of all activities, because the externalities of research are so large (Jones and Williams, 1998; Bloom et al., 2013; Gross and

[28] In the nature-leaning and nurture-leaning scenarios, genetic transmission explains respectively 60% and 7% of the empirical inequality. The mainline parameterization represents our best guesses at the true values, which are broadly supported by all of the major AI reasoning models (Appendix Table C2), but, as noted above, we have wide priors on some of the parameters.

Sampat, 2023). Second, talent is multi-dimensional; the individuals who would be most successful in the sciences would be unlikely to find equal levels of success in other domains (Murphy et al., 1991). Third, the unequal barriers that prevent advancement in the sciences also exist in other domains. For example, authors of highly-cited patents are ten times more likely to come from the top 5% of the father income distribution than from the bottom 50% (Bell et al., 2019); nationality and gender play a role in this inequality as well. Statistics are similar for doctors, CEOs of large companies, and politicians (Dal Bó et al., 2017; Thompson et al., 2019). The evidence suggests that talented individuals who face social barriers to success in the sciences are unlikely to find easy paths in other domains either.

Second, are "missing Nobel laureates" in fact generating important scientific discoveries, but simply not being recognized for their work? The physicist Lise Meitner was nominated for the Nobel Prize more than 40 times for her work on nuclear fission but was overlooked when the prize was given to her colleague Otto Hahn. We are doubtful that stories like this can explain a large share of our missing scientists, because gatekeeping in the sciences begins long before Nobel prizes are being considered. To reach the top of the sciences, an individual must pass through many filters, each of which may disadvantage people from less affluent backgrounds. Entry into elite PhD programs and postdocs, grants, and early career awards are all key ingredients of success, and all of these are based on evaluation by peer scientists, like the Nobel Prize itself (Merton, 1968; Stephan, 2012). It is unlikely that the unequal outcomes we describe here are driven entirely by unequal selection only at the final stage of one's scientific career; it is more likely the result of an accumulation of inequalities over the life cycle. As such, the failure points of missing scientists are more likely to be spread throughout the career pipeline than concentrated at the final stage.

5 Conclusion

When opportunity is unequal, great talents may be under-nourished and potential scientific discoveries could be lost. Nobel laureates in the sciences overwhelmingly originate from the top tiers of national income distributions, and are even more elite in the global income distribution. The small

number of female winners and their even more elite backgrounds also suggest substantial untapped potential.

Comparison with estimates from the prior literature shows that more successful scientists also come from better off families than less successful scientists. In concurrent work, Abramitzky et al. (2024) estimate that U.S. academics at U.S. universities in 1900–69 had an average father SES rank of 72.2. Airoldi and Moser (2024) report the median U.S. scientist in 1921 had a 78th percentile father, with the median "star" scientist (rated among the 15% most successful by their peers) ranking slightly higher. By contrast, our results show that the average Nobel laureate in that era had a father rank of 89.

Social and economic changes over the 20th century made it possible for a growing share of talented people from lower income classes (in rich countries at least) to make it to the top of the sciences and contribute their talents to humanity. Progress at bridging global opportunity gaps has been much smaller. Our results suggest that there is a large reservoir of untapped scientific talent—in lower-income countries in particular—that is missing the complementary inputs for success, which could limit human progress. Improving access to those complementary inputs is desirable for the sake of fairness alone, but could also unleash new avenues of innovation, growth, and advancement for humankind.

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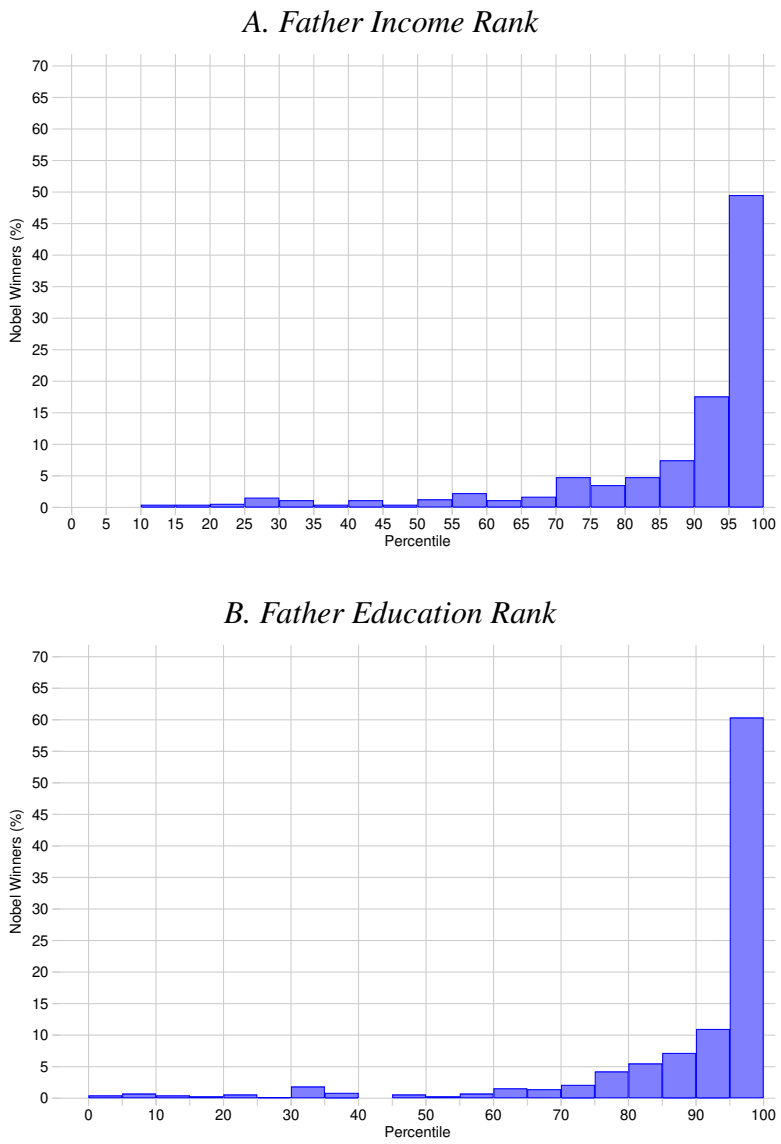
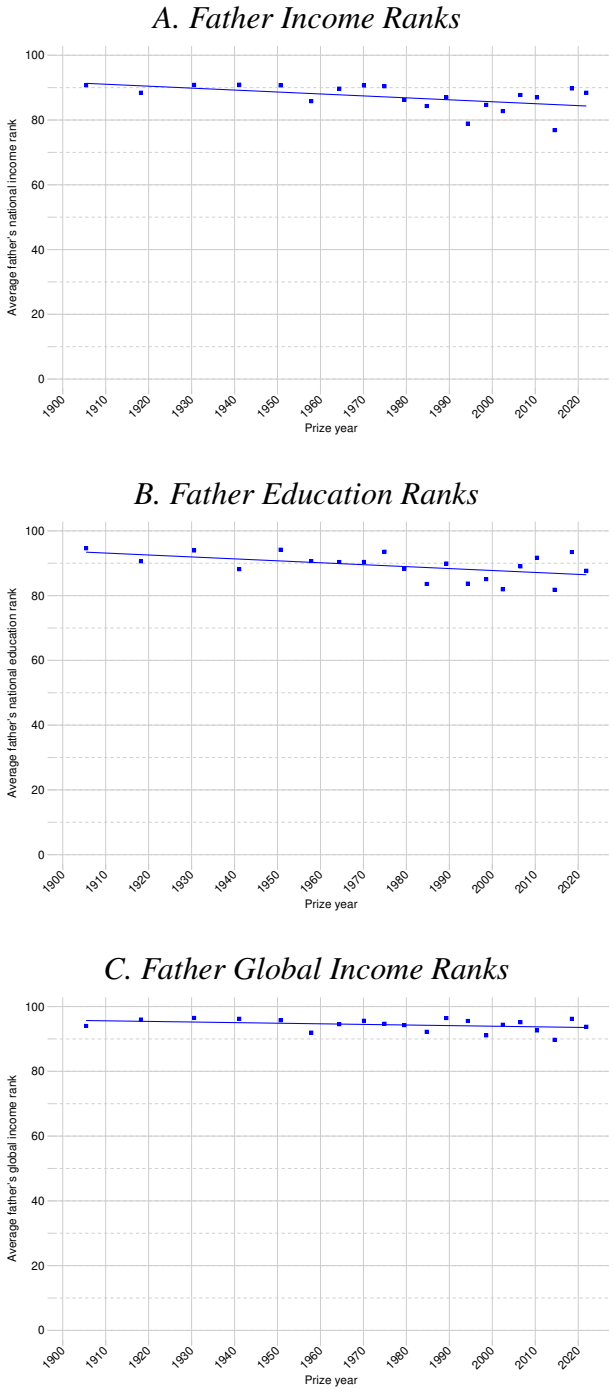
Figures and Tables

Figure 1: Distribution of Socioeconomic Ranks of Fathers of Nobel Laureates



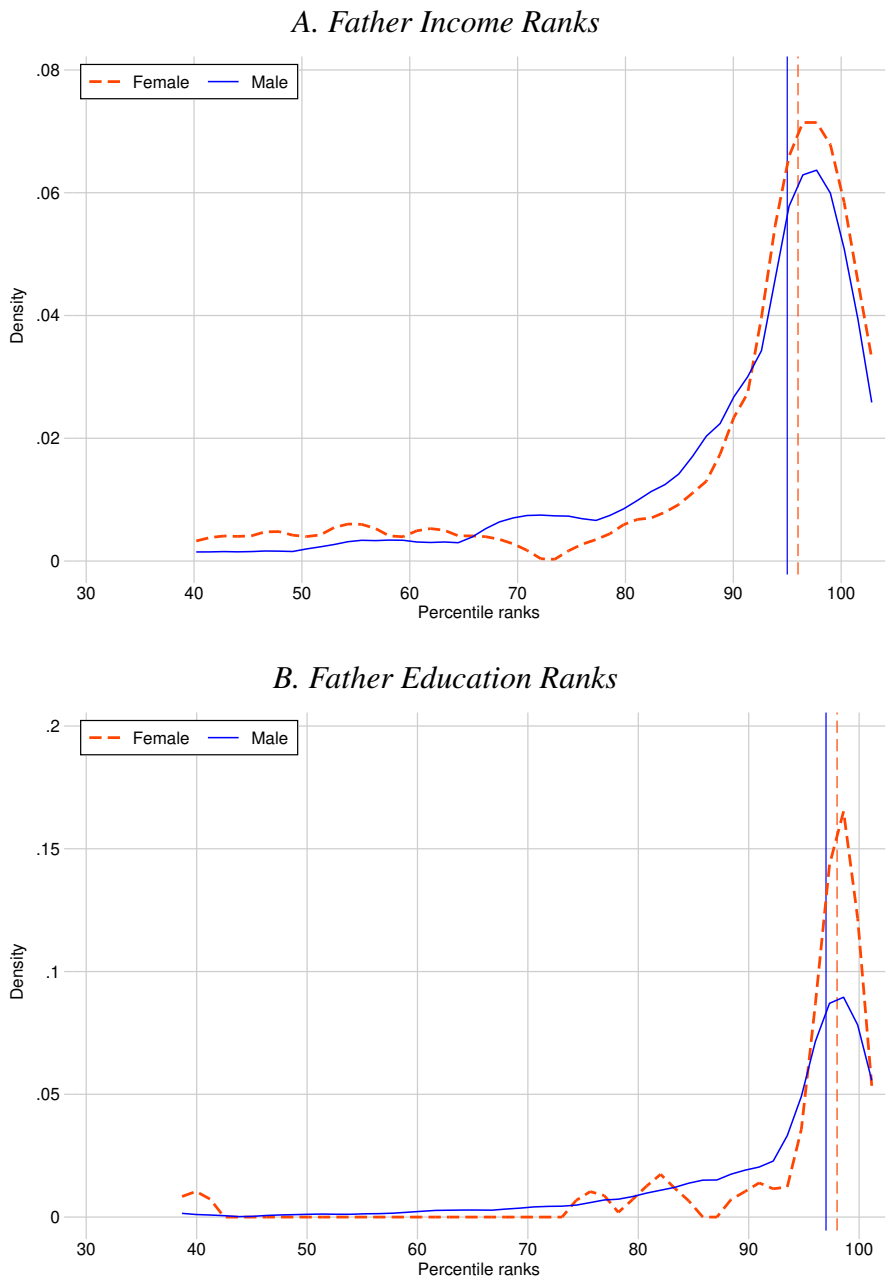
Notes: The graph shows the frequency distribution of national father income and education percentiles (bin size = 5) for all Nobel laureates in the sciences in our sample. Father income and education percentiles were predicted from father occupations, using data from the 1940 and 1950 U.S. Population Censuses.

Figure 2: Average Socioeconomic Ranks of Fathers of Nobel Laureates



Notes: The graph shows binned scatterplots of national father's income and education rankings, and global income rankings for Nobel laureates in the sciences from 1901–2023. Father income and education percentiles were predicted from father occupations, using data from the 1940 and 1950 U.S. Population Censuses. The fitted lines are estimated using ordinary least squares (OLS).

Figure 3: Socioeconomic Rank Distribution of Fathers of Nobel Laureates, by Laureate Gender



Notes: The graphs show the kernel density functions of national socioeconomic rankings of fathers of all Nobel laureates in the sciences, by laureate gender. The vertical lines represents the median percentile within each gender. Father income (Panel A) and education (Panel B) percentiles were predicted from father occupations, using data from the 1940 and 1950 U.S. Population Censuses.

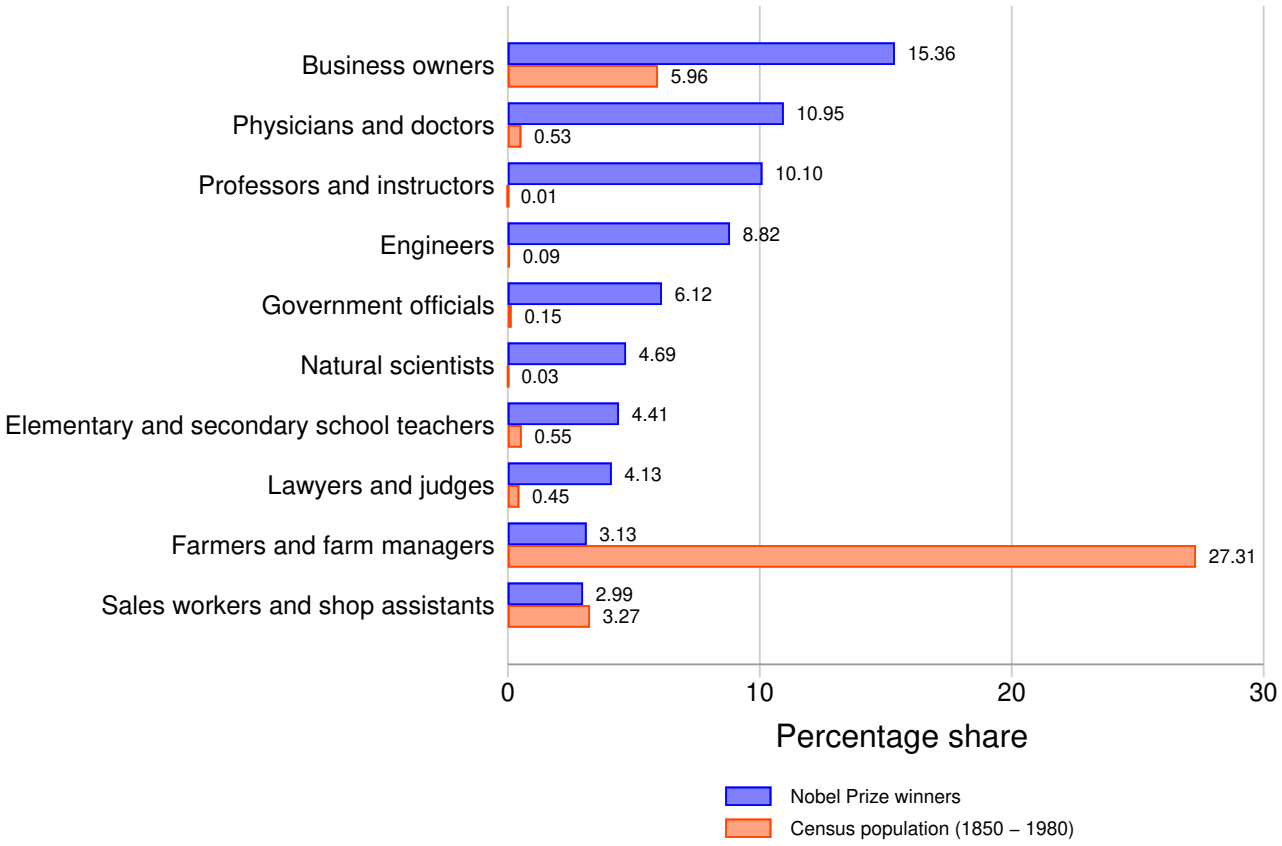
Table 1: Heterogeneity in Socioeconomic Rank of Fathers of Nobel Laureates

	Average Rank		Top 10% Indicator		Top 5% Indicator		
	National Income	National Education	National Income	National Education	National Income	National Education	*N*
Full Sample	87.081 (0.677)	89.187 (0.679)	0.671 (0.018)	0.713 (0.017)	0.495 (0.019)	0.603 (0.018)	711
Panel A: Gender							
Male	86.977 (0.695)	89.004 (0.701)	0.668 (0.018)	0.709 (0.017)	0.492 (0.019)	0.596 (0.019)	683
Female	89.600 (2.900)	93.666 (2.261)	0.750 (0.082)	0.821 (0.072)	0.571 (0.094)	0.786 (0.078)	28
F-test	[0.379]	[0.049]	[0.327]	[0.131]	[0.406]	[0.018]	
Panel B: Prize Category							
Chemistry	86.451 (1.329)	87.864 (1.331)	0.670 (0.035)	0.654 (0.033)	0.481 (0.037)	0.541 (0.036)	185
Physics	87.554 (1.245)	91.104 (1.247)	0.687 (0.032)	0.758 (0.031)	0.502 (0.034)	0.649 (0.034)	211
Medicine	87.585 (1.211)	88.897 (1.213)	0.691 (0.031)	0.726 (0.030)	0.534 (0.033)	0.650 (0.033)	223
Economics	86.038 (1.885)	88.156 (1.888)	0.587 (0.049)	0.696 (0.047)	0.413 (0.052)	0.511 (0.051)	92
F-test	[0.838]	[0.296]	[0.311]	[0.134]	[0.263]	[0.016]	
Panel C: Regions							
USA	83.829 (1.120)	86.597 (1.128)	0.584 (0.029)	0.639 (0.028)	0.396 (0.031)	0.525 (0.030)	255
Western Europe	88.743 (1.003)	90.570 (1.010)	0.701 (0.026)	0.736 (0.025)	0.531 (0.028)	0.632 (0.027)	318
Eastern Europe	92.801 (2.412)	94.753 (2.428)	0.818 (0.063)	0.891 (0.060)	0.727 (0.067)	0.727 (0.066)	55
Global South	87.739 (3.509)	88.398 (3.532)	0.769 (0.091)	0.846 (0.088)	0.538 (0.097)	0.692 (0.095)	26
All others	86.528 (2.370)	88.048 (2.385)	0.702 (0.062)	0.684 (0.059)	0.491 (0.065)	0.632 (0.064)	57
F-test	[0.002]	[0.013]	[0.002]	[0.001]	[0.000]	[0.016]	

Notes: The table shows group means for measures of Nobel laureate fathers' socioeconomic status in different subsamples. Standard errors are in parentheses. The first two columns show the average national income and education rank respectively, calculated as above. Columns 3 and 4 show the share of laureates in the given category who are classified in the top 10% of the income and education distributions, respectively. Columns 5 and 6 show the same for the top 5%. P-values from the F-test of equality of all group means are displayed below each set of means. Appendix Table A10 lists the countries in each region.

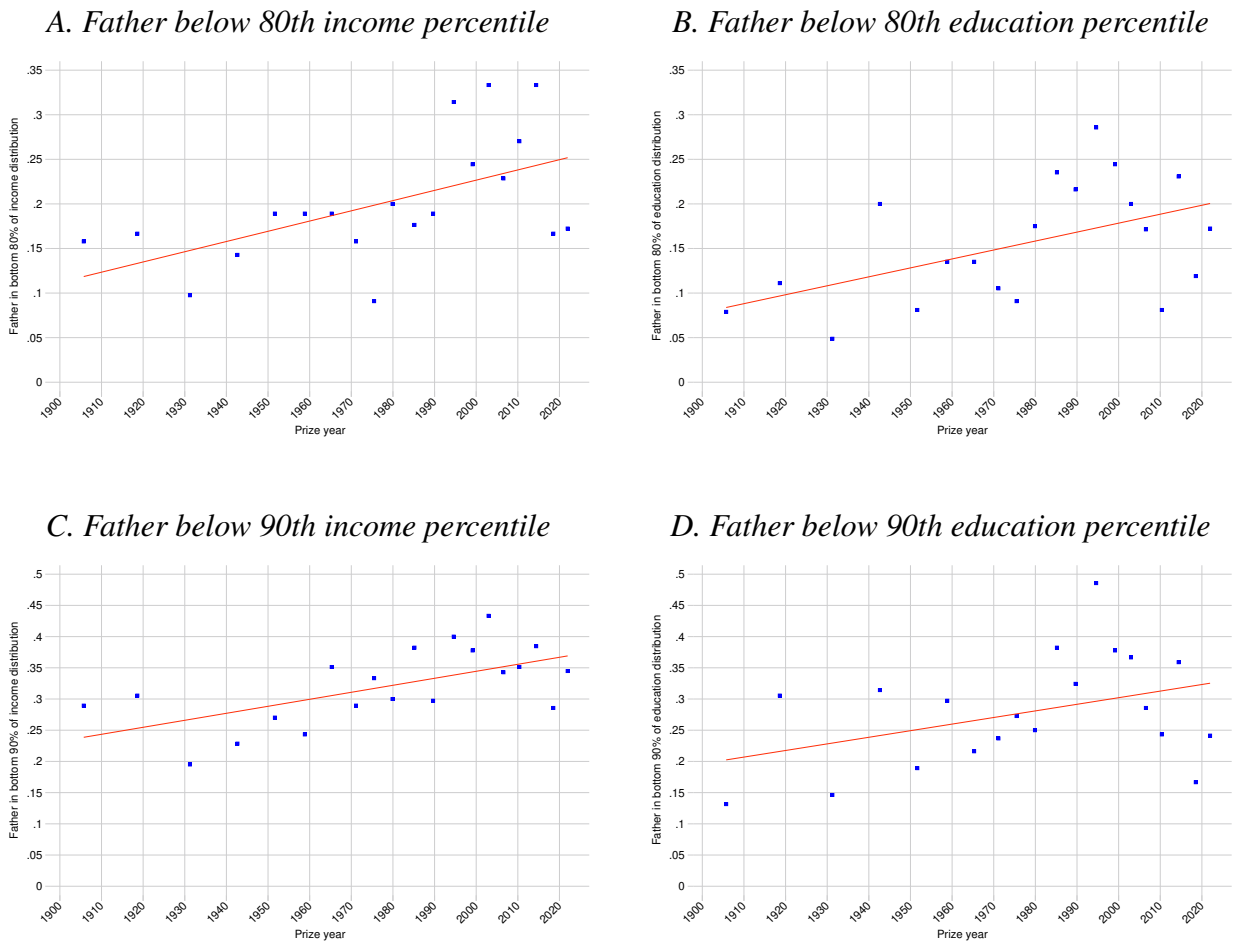
A Additional Figures and Tables

Figure A1: Most Frequent Father Occupations among Nobel Laureates



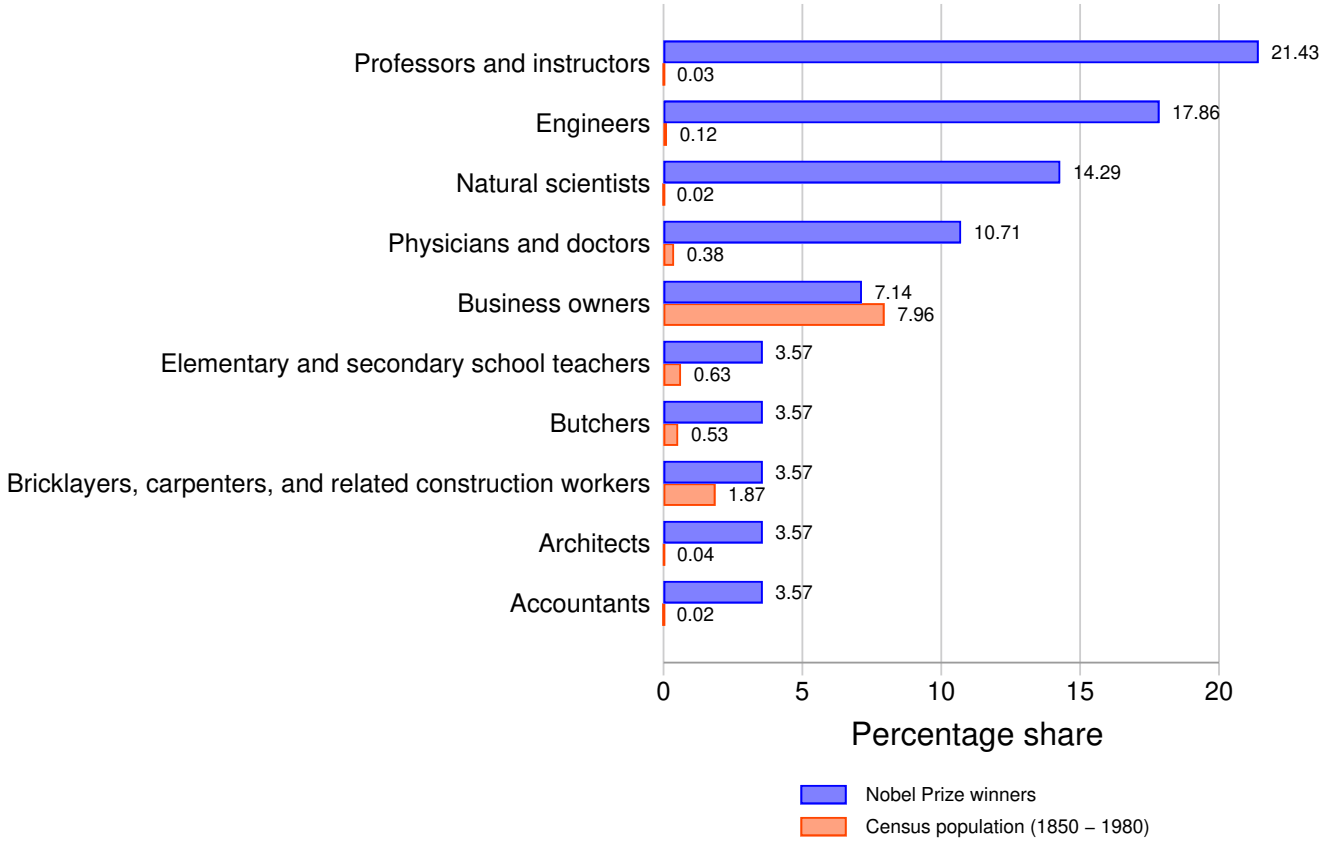
Notes: The graph shows the distribution of father occupations among Nobel laureates in the sciences (in blue) and the general population (in orange). Occupational shares are averages from the U.S. Population Census between 1850 and 1980. The laureate sample covers the full set of prizes from 1901–2023.

Figure A2: Share of laureates with non-elite fathers



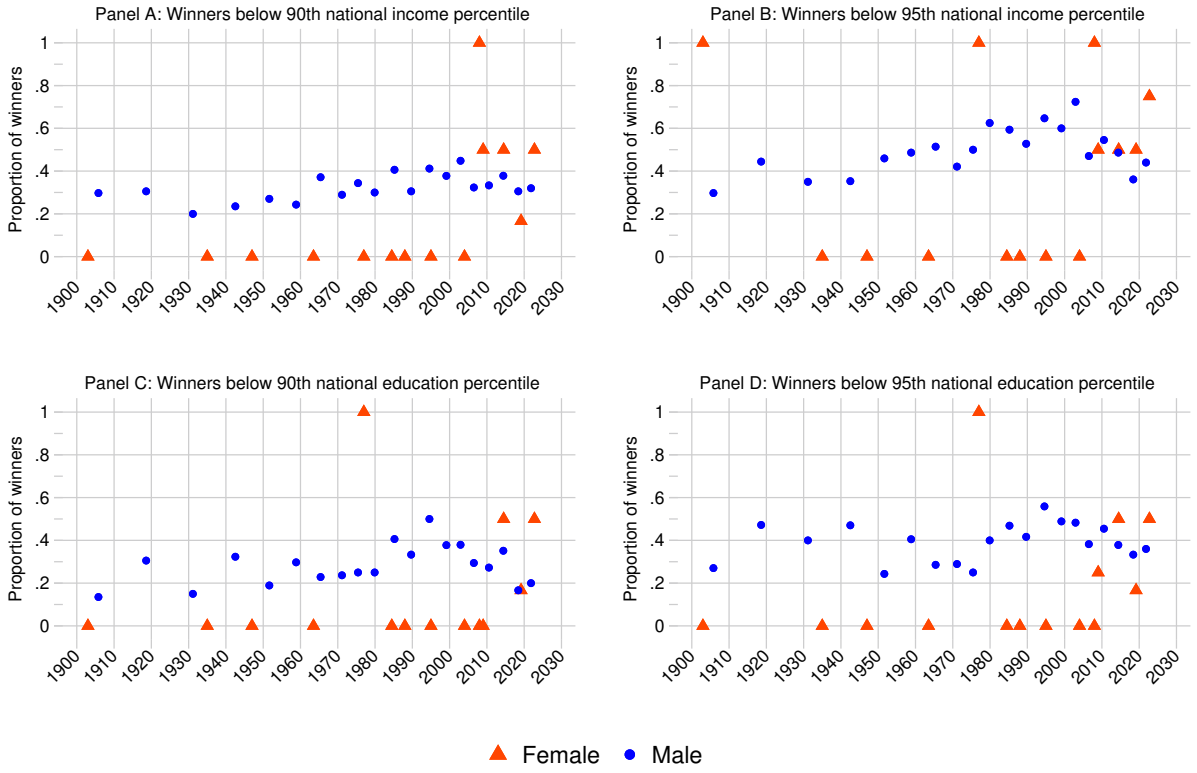
Notes: The graphs show binned scatterplots of the share of Nobel laureates whose fathers were below the 80th percentile (Panels A and B) or 90th percentile (Panels C and D) of the income (Panels A and C) and education (Panels B and D) distributions. Father income and education percentiles were predicted from father occupations, using data from the 1940 and 1950 U.S. Population Censuses.

Figure A3: Most Frequent Father Occupations among Female Nobel Laureates



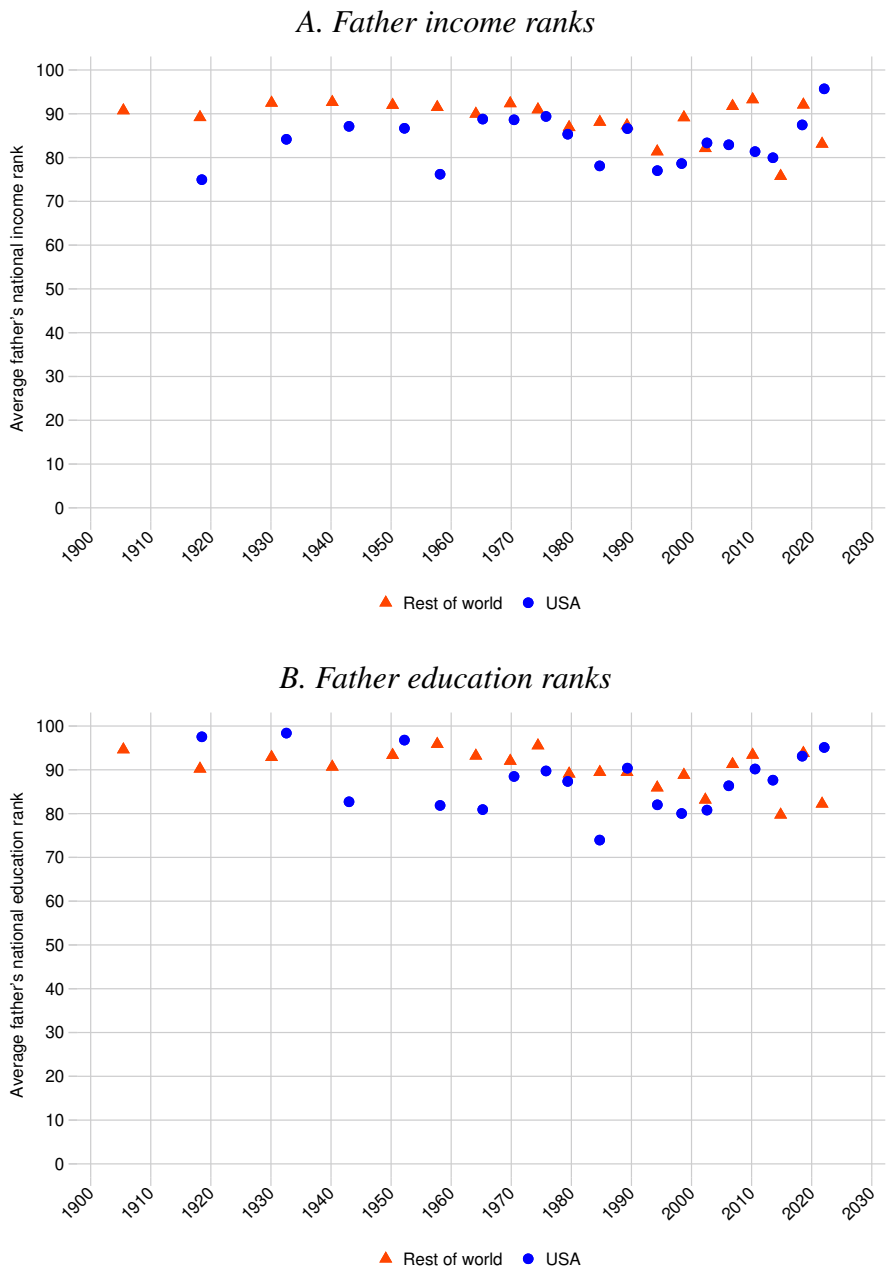
Notes: The graph shows the distribution of father occupations among *female* Nobel laureates in the sciences (in blue) and the general population (in orange). Occupational shares are averages from the U.S. Population Census between 1850 and 1980. The laureate sample covers the full set of prizes from 1901–2023.

Figure A4: Share of Nobel Laureates with Fathers below a given SES Rank



Notes: The graph shows binned scatterplots of the proportion of winners in each time period with fathers below a given income and education percentile. Circles represent male laureates and triangles female laureates. Father income and education percentiles were predicted from father occupations, using data from the 1940 and 1950 U.S. Population Censuses.

Figure A5: Average SES Ranks of Fathers of Nobel Laureates: USA vs. Rest of World



Notes: The graph shows binned scatterplots of the proportion of winners in each time period with fathers below a given income (Panel A) and education (Panel B) percentile. Circles represent laureates who grew up in the United States, and triangles represent winners who grew up elsewhere. Father income and education percentiles were predicted from father occupations, using data from the 1940 and 1950 U.S. Population Censuses.

Table A1: Most Common Father Occupations of Nobel Laureates, by Prize Category

Father Occupation	Prize Category			
	Chemistry	Physics	Medicine	Economics
Business Owners	19%	14%	13%	15%
Physicians and doctors	9%	8%	15%	11%
Professors and academics	10%	14%	9%	5%
Engineers	8%	11%	8%	8%
Natural Scientists	3%	6%	4%	7%
Lawyers and Judges	4%	2%	5%	4%
Government Officials	5%	5%	7%	8%
Teachers	4%	7%	2%	5%
Retail Workers	3%	3%	3%	2%
Farmers	4%	2%	3%	4%
Observations	185	211	223	92

Notes: The table shows the ten most common occupations among fathers of the Nobel laureates in our sample, by prize category.

Table A2: Secular Trends in Average SES Ranks of Fathers of Nobel Laureates

	National Income Rank	National Education Rank
Constant (Year = 1901)	91.596***	93.687***
	(1.445)	(1.268)
Prize Year	-0.060***	-0.060***
	(0.019)	(0.018)
Observations	711	711

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table shows estimates from OLS regressions of father socioeconomic rank on prize year. The prize year variable has been rescaled to a start date of zero (representing 1901), so the constant can be interpreted as the fitted value for the first year of our sample. Standard errors are reported in parentheses. Father income and education percentiles were predicted from father occupations, using data from the 1940 and 1950 U.S. Population Censuses.

Table A3: Average SES Ranks of Fathers of Nobel Laureates and Time Trends: Robustness

	National Income Rank	National Education Rank
Panel A: Sample excluding difficult to classify observations		
Constant (Year = 1901)	91.720	94.136
	(1.467)	(1.223)
Prize Year	-0.060***	-0.064***
	(0.019)	(0.017)
Observations	698	698
Panel B: Sample excluding aristocrats		
Constant (Year = 1901)	91.104	93.304
	(1.491)	(1.312)
Prize Year	-0.055***	-0.056***
	(0.019)	(0.018)
Observations	703	703
Panel C: Sample excluding business owners and farmers		
Constant (Year = 1901)	92.320	95.719
	(1.372)	(1.120)
Prize Year	-0.052***	-0.049***
	(0.018)	(0.016)
Observations	580	580
Panel D: Sample excluding economics		
Constant (Year = 1901)	91.688	93.924
	(1.474)	(1.291)
Prize Year	-0.062***	-0.064***
	(0.020)	(0.019)
Observations	619	619

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table shows estimates from an OLS regression of Nobel laureate father SES rank on a linear time trend. These specifications are based on the primary specification in Appendix Table A2, with the following modifications. In Panel A, we exclude 13 laureates whose father occupations were vaguely described and difficult to classify. In Panel B, we exclude 8 laureates whose fathers were aristocrats (who were classified as being at education and income percentile 99 in the main analysis). In Panel C, we exclude 130 children of business owners and farmers, since these occupations can be associated with a wide range of possible SES ranks. In Panel D, we exclude the Nobel Prize in Economics.

Table A4: Average SES Ranks of Fathers of Nobel Laureates and Time Trends: Robustness to Alternate Codings

	National Income Rank	National Education Rank
Panel A: Heterogeneous coding of business owners		
Constant (Year = 1901)	91.440	93.261
	(1.462)	(1.319)
Prize Year	-0.063***	-0.065***
	(0.019)	(0.018)
Observations	711	711
Panel B: Recoding education according to biographies		
Constant (Year = 1901)		93.435
		(1.292)
Prize Year		-0.061***
		(0.018)
Observations		711
Panel C: Recoding education and heterogeneous business owners		
Constant (Year = 1901)		93.435
		(1.292)
Prize Year		-0.061***
		(0.018)
Observations		711

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table shows estimates from an OLS regression of Nobel laureate father SES rank on a linear time trend. These specifications are based on the primary specification in Appendix Table A2, with the following modifications. In Panel A, we classified business owners as small, medium or large. For small business owners, we used the 25th percentile education and income data from the Census to calculate the father's rank. For large, we used the 75th. In Panel B, we override the father's education rank in cases where the historical record documents the actual education level of the father. In Panel C, we do both.

Table A5: Differences in Average SES Rank of Laureate Fathers, by Gender: Robustness

	Average Rank		Top 10% Indicator		Top 5% Indicator	
	National Income Rank	National Education Rank	National Income Rank	National Education Rank	National Income Rank	National Education Rank
Panel A: Sample excluding difficult to classify observations						
Constant (Year = 1901)	87.027	89.127	0.669	0.709	0.498	0.595
	(0.702)	(0.698)	(0.018)	(0.018)	(0.019)	(0.019)
Female Winners	3.985	4.573	0.109	0.105	0.095	0.220***
	(2.732)	(2.447)	(0.082)	(0.077)	(0.097)	(0.077)
Observations	698	698	698	698	698	698
Panel B: Sample excluding aristocrats						
Constant (Year = 1901)	86.823	88.874	0.664	0.705	0.486	0.591
	(0.701)	(0.708)	(0.018)	(0.018)	(0.019)	(0.019)
Female Winners	2.777	4.792	0.086	0.116	0.086	0.195**
	(2.984)	(2.370)	(0.084)	(0.075)	(0.096)	(0.080)
Observations	703	703	703	703	703	703
Panel C: Sample excluding business owners and farmers						
Constant (Year = 1901)	85.825	89.433	0.635	0.801	0.540	0.735
	(0.813)	(0.824)	(0.020)	(0.017)	(0.021)	(0.019)
Female Winners	3.732	5.326	0.134	0.083	0.076	0.111
	(3.227)	(2.435)	(0.085)	(0.065)	(0.098)	(0.073)
Observations	603	603	580	580	580	580
Panel D: Sample excluding economics						
Constant (Year = 1901)	86.823	88.874	0.677	0.710	0.502	0.608
	(0.747)	(0.753)	(0.019)	(0.019)	(0.021)	(0.020)
Female Winners	4.281	5.051	0.163**	0.130*	0.138	0.232***
	(2.936)	(2.532)	(0.076)	(0.076)	(0.098)	(0.076)
Observations	619	619	619	619	619	619

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table shows estimates from an OLS regression of Nobel laureate father SES rank on an indicator for female laureates. The specifications are based on those in Panel A of Table 1, with the following modifications. In Panel A, we exclude 13 laureates whose father occupations were vaguely described and difficult to classify. In Panel B, we exclude eight laureates whose fathers were aristocrats (who were classified as being at education and income percentile 99 in the main analysis). In Panel C, we exclude 130 children of business owners and farmers, since these occupations can be associated with a wide range of possible SES ranks. In Panel D, we exclude the Nobel Prize in Economics.

Table A6: Differences in Average SES Rank of Laureate Fathers, by Gender: Robustness to Alternate Codings

	Average Rank		Top 10% Indicator		Top 5% Indicator	
	National Income Rank	National Education Rank	National Income Rank	National Education Rank	National Income Rank	National Education Rank
Panel A: Heterogeneous coding of business owners						
Constant (Year = 1901)	86.688	88.364	0.654	0.714	0.499	0.616
	(0.717)	(0.770)	(0.018)	(0.017)	(0.019)	(0.019)
Female Winners	2.815	2.706	0.060	0.107	0.072	0.169**
	(3.107)	(3.855)	(0.087)	(0.075)	(0.096)	(0.080)
Observations	698	698	711	711	711	711
Panel B: Recoding education according to biographies						
Constant (Year = 1901)		88.959		0.710		0.593
		(0.712)		(0.017)		(0.019)
Female Winners		2.631		0.076		0.193**
		(3.461)		(0.080)		(0.080)
Observations		698		711		711
Panel C: Recoding education and heterogeneous business owners						
Constant (Year = 1901)		88.959		0.710		0.593
		(0.712)		(0.017)		(0.019)
Female Winners		2.631		0.076		0.193**
		(3.461)		(0.080)		(0.080)
Observations		698		711		711

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table shows estimates from an OLS regression of Nobel laureate father SES rank on an indicator for female laureates. The specifications are based on those in Panel A of Table 1, with the following modifications. In Panel A, we classified business owners as small, medium or large. For small business owners, we used the 25th percentile education and income data from the Census to calculate the father's rank. For large, we used the 75th. In Panel B, we override the father's education rank in cases where the historical record documents the actual education level of the father. In Panel C, we do both.

Table A7: Gender Differences in Laureate Father Ranks Before and After 2000

	Average Rank		**Top 10% Indicator**		**Top 5% Indicator**	
	National Income Rank	National Education Rank	National Income Rank	National Education Rank	National Income Rank	National Education Rank
Pre-2000	87.605	89.685	0.687	0.712	0.503	0.605
	(0.800)	(0.781)	(0.021)	(0.021)	(0.023)	(0.022)
Female Winners	9.719	7.334	0.313***	0.188*	0.297**	0.295***
	(1.299)	(1.707)	(0.021)	(0.097)	(0.129)	(0.098)
Post-2000	-2.210	-2.396	-0.069	-0.011	-0.039	-0.033
	(1.599)	(1.680)	(0.041)	(0.039)	(0.042)	(0.042)
Post-2000 x Female Winners	-9.805	-2.820	-0.320***	-0.112	-0.316*	-0.145
	(4.562)	(4.035)	(0.122)	(0.142)	(0.178)	(0.148)
Observations	711	711	711	711	711	711

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table reports estimates from OLS regressions of father SES rank on a female laureate indicator, interacted with an indicator for Nobel prizes awarded after 2000. The "female winner" shows that female winners consistently came from more elite families for prizes awarded before 2000. The point estimate of the interaction term is negative, suggesting that this effect may have diminished in recent years, but the term is statistically significant only in one out of six specifications. There are 10 female winners in our sample in the first 100 prize years (1901–1999), and 18 from 2000–2023.

Table A8: Secular Trends in Father SES Ranks of Nobel Laureates, by Prize Category

	National Income Rank	National Education Rank
Panel A: Chemistry		
Constant (Year = 1901)	93.875	94.849
	(2.181)	(1.956)
Prize Year	-0.102***	-0.096***
	(0.034)	(0.034)
Observations	185	185
Panel B: Economics		
Constant (Year = 1901)	89.716	85.765
	(11.614)	(11.525)
Prize Year	-0.038	0.024
	(0.118)	(0.116)
Observations	92	92
Panel C: Medicine		
Constant (Year = 1901)	93.720	94.832
	(2.721)	(2.488)
Prize Year	-0.086**	-0.084**
	(0.037)	(0.035)
Observations	223	223
Panel D: Physics		
Constant (Year = 1901)	87.906	92.335
	(2.586)	(2.119)
Prize Year	-0.005	-0.017
	(0.034)	(0.030)
Observations	211	211

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table shows OLS regression estimates of laureate father SES rank on prize year, separately for each prize category. The prize year variable has been rescaled to a start date of zero, so the constant is interpreted as the fitted value for the father ranks of laureates in 1901. Standard errors are in parentheses.

Table A9: Father Ranks of Nobel Laureates: USA vs. Rest of World

	Average Rank		**Top 10% Indicator**		**Top 5% Indicator**	
	National Income Rank	National Education Rank	National Income Rank	National Education Rank	National Income Rank	National Education Rank
Constant	88.899	90.636	0.719	0.754	0.550	0.647
	(0.792)	(0.762)	(0.021)	(0.020)	(0.023)	(0.022)
US-born Winners	-5.069	-4.038	-0.135***	-0.115***	-0.154***	-0.121***
	(1.458)	(1.509)	(0.037)	(0.036)	(0.039)	(0.039)
Observations	711	711	711	711	711	711

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table shows estimates from OLS regressions of SES ranks of fathers of Nobel laureates on an indicator that takes the value one for laureates who spent their childhood in the United States. Standard errors are reported in parentheses.

Table A10: Countries in each region

Western Europe	Eastern Europe	Global South	Others
Austria	Azerbaijan	Algeria	Australia
Belgium	Belarus	Argentina	Canada
Cyprus	Bosnia and Herzegovina	Brazil	Israel
Denmark	Croatia	China	Japan
Finland	Czech Republic	Egypt	New Zealand
France	Hungary	India	Turkey
Germany	Latvia	Indonesia	
Ireland	Lithuania	Lebanon	
Italy	Poland	Mexico	
Luxembourg	Romania	Morocco	
Netherlands	Russia	Pakistan	
Norway	Slovakia	South Africa	
Portugal	Slovenia	South Korea	
Spain	Ukraine	Taiwan	
Sweden		Venezuela	
Switzerland			
United Kingdom			

Notes: The table lists the countries included in each of the geographical regions titled "Western Europe", "Eastern Europe", "Global South", and "All others" that are used in the analysis for Table 1.

Table A11: Global Rank Estimates: Robustness to National Occupation Distribution Adjustment

	Global Rank (Original)	Global Rank (No Occupation Adjustment)
Constant (Year = 1901)	95.771***	95.330***
	(0.708)	(0.782)
Prize Year	-0.018**	-0.022**
	(0.009)	(0.011)
Observations	711	703

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table shows estimates from a regression of laureate global ranking on a constant and on the number of years since the first Nobel Prize in 1901. Column 1 corresponds to the primary estimates displayed in Panel C of Figure 2. In Column 2, global income ranks are calculated *without* adjusting national occupation rank distributions for country GDP level. Standard errors are in parentheses.

B Additional Information on Data and Methods

B.1 Biographical Data

We hand-collected biographical information for every Nobel laureate in Physics, Chemistry, Physiology, and Economics. For each laureate, we reviewed multiple sources to identify their birth dates, birth locations, and the occupations of their parents. The availability of parent occupational information was heterogeneous. Many winners described their parents work in their acceptance speeches or biographies. Parent occupations were also recorded in documents like obituaries, biographies, scientific directories and encyclopedia entries. When we could not find information on father occupation, we reached out to living laureates, a handful of whom responded with information about their parents. We obtained data on father occupations for 711 out of 739 laureates (96.2%), but only 181 mother occupations (25.3%). Mothers' occupations are also generally less informative about laureates' childhood socioeconomic status, because so many women in this era were out of the labor force.

When laureates moved before the age of six, we assigned them to their location at age six. Results were unchanged if we used other ages as anchors for their home country. When laureates were born to expatriates (i.e. parents living temporarily in countries other than their home country), we assigned them to their parents' home countries, as the home country occupational distributions are likely to be more relevant for determining parental rank. We similarly assigned them to home countries when parents were serving in a foreign service. For example, Ronald Ross (Physiology, 1902) was the son of a British general serving in India; at age 8, he moved to the Isle of Wight for school. Our assumption is that his parents' socioeconomic status is better characterized by the average status of an army general in the United Kingdom than that of an army general in India. There are only a handful of cases like these, so they do not have substantial bearings on our results.

We construct cohorts of winners by rounding each winner's birth year to the nearest decade (e.g., a winner born in 1893 belongs to the 1890 cohort, whereas a winner born in 1898 belongs to the 1900 cohort). There are six winners born between 1835 - 1843, all of whom are initially

assigned to the 1840 cohort. However, due to the lack of availability of census data during these years, we then map these observations to 1850 instead, the closest available decade.

B.2 Occupational Earnings and Education from U.S. Population Census

To estimate the socioeconomic status of laureates' parents, we aimed to identify the average income and education ranks of people with the same occupation as each laureate's father in the U.S. Census. We used data from the 1850–1970 U.S. population censuses, extracted from IPUMS and proceeded as follows.

We restricted the census sample to working age (18–65 years old) men, and dropped all individuals who were not in the labor force. We also dropped census observations for which the occupational category variable takes on the following values: "students", "disabled individuals with no reported occupation", "other non-occupation", "n/a (blank)". Excluding the unemployed could marginally bias our ranks downward, but the bias should be small, given that (i) the historic U.S. unemployment rate has typically fluctuated between only 5–8%; and (ii) the unemployed are distributed across many different occupations, limiting the degree of bias.

We matched each laureate's father occupation to the nearest harmonized occupational category (*occ1950*, created by IPUMS). When fathers held multiple occupations, we assigned the one described as the primary occupation during the laureate's childhood years. Appendix Table B1 shows the list of occupational categories used in the sample.

There is inevitably some measurement error in classifying business owners, as the historical record was often unclear on the scale of the business; as such, a business owner could represent a relatively upper or middle class family. We classified business owners with code 290, which is "Managers, officials, and proprietors, not otherwise classified." This puts business owners, on average, at the 86th education percentile and the 93rd income percentile. Farm owners exist as an occ1950 category, but similarly cover a wide range of socioeconomic ranks. To ensure that our assumptions about these occupational ranks were not driving any of our results, we verified that our primary results are similar when we exclude business owners and farmers (Appendix Tables A3 and

A5).

Individual and household earnings were available beginning in 1950, and years of education in 1940, so we used these values for all earlier years. We excluded negative or zero earnings values. Literacy is recorded in earlier census years, but is considered less reliable by historians, as collection methodology varied substantially across enumerators (Collins and Margo, 2006). For census years after 1940/50, we applied a three-census moving average to smooth out fluctuations in occupational income/educations from sampling variation and interpolated missing values.[29]

We also estimated the population share of each occupation in each decade, which is essential for calculating ranks. To calculate ranks, we assigned individuals an earnings level based on their occupation, and then ordered individuals within each decadal census according to those earnings levels. We then assigned the mean earnings rank within each occupation group, and collapsed the data to occupation-decade level. These earnings ranks were then matched to the Nobel laureates database at the occupation-decade level. We performed a similar exercise for education ranks.

Since occupational incomes were not observed before 1950, there is no change in the relative ordering of occupations before 1950. However, we observe the share of people with each occupation in all census years, so the *ranks* of occupations do change continuously over time. For instance, trade workers tended to have relatively high education ranks in 1850, but had lower ranks by 1950, due to the growing number of more educated white collar workers.

B.3 Historical Estimates of Country-Level GDP per Capita and Population

We used country of childhood in two ways in our analysis. For our primary analysis, we used country of childhood to adjust the relative occupational distributions, to reflect the fact that an individual with only some schooling (say, literacy) would occupy a higher education rank in a poor country than in a rich country. Second, for the section on income differences across countries only, we used national GDP per capita to adjust occupational earnings for living standards across countries. This second adjustment reflects the fact that an individual at the 90th percentile of the

[29]These adjustments have little bearing on our results, as they only affect census years after 1940/50, and most Nobel laureates were born earlier.

earnings distribution in India has a much lower global income rank than an individual at the 90th percentile in the United States.

We used historical estimates of GDP per capita and population for 217 countries from 1830–1970, from Fariss et al. (2022), with adjustments described below.

B.3.1 Adjusting for Occupation Distributions

The distribution of occupational population shares affects the resulting socioeconomic rank of each occupation, even if the ordering of occupations is unchanged. For example, the percentile rank of a sawyer in the U.S. is much higher in 1850, a time when roughly 44% of the population were farmers, compared to 1950, when that share is down to only 9%. Similarly, countries will have different occupational distributions at different levels of economic development. However, we only had data on the historical occupational distribution of the United States.

We therefore used the U.S. historical occupational distribution to proxy for the occupational distribution in other countries at different points in time. We assigned each country the U.S. occupational distribution from the decade when its development level (measured by GDP per capita) was nearest to that of the U.S., using minimum absolute distance.

Consider an example of how we would calculate the earnings rank of an Indian tailor in 1950. First, we identified the U.S. decade in our sample data with the nearest GDP per capita level to that of India in 1950 — this was 1850. In 1850 in the U.S., a tailor had an earnings rank of 84; we assign this rank to the Indian tailor for the primary analysis. By contrast, in the U.S. in 1950, a tailor would have an earnings rank of only 55.

B.3.2 Calculating Global Income Ranks

Our primary analysis focuses on occupational ranks within countries. Since most winners are from the West, this tells us how effectively the West is at mobilizing its own talent. In the final part of the results, we instead consider global income ranks, which tell us how effective human society is at mobilizing global talent. Consider again the Indian tailor in 1950 above. The tailor occupied the 84th earnings rank in India, but their global earnings rank was much lower, since India in 1950 was

a very poor country.

We calculate a global earnings rank as follows. First, for each country-decade-occupation group, we scale the occupational income by the GDP per capita ratio of each country with the United States in the same decade. This gives us a real income estimate for each country-decade-occupation group. Second, we create a synthetic global population based on each country's occupation shares (adjusted for development level, as above) and their population. An individual's rank in this synthetic population is that individual's global income rank.

Let us continue with the example of the Indian tailor above. We have already estimated that the tailor is at the 84th earnings percentile in India in 1950. In 1950, U.S. GDP per capita was approximately 16 times higher than Indian GDP; we therefore multiply the Indian tailor's predicted income by 0.06. After conducting this calculation for every country-occupation-decade, the Indian tailor has a global income rank of 45. This reflects the fact that the tailor is relatively well off in India, but India is relatively poor globally. Appendix Figure B1 summarizes the example.

For individuals from poor countries, global ranks are systematically lower than domestic ranks, and vice versa. A U.S. tailor in 1950 would be at the 97th percentile globally. Since most of the Nobel laureates are from rich countries, the average global earnings rank is therefore higher than the average domestic earnings rank.

Figure B1: Example Calculation of National and Global Earnings Ranks

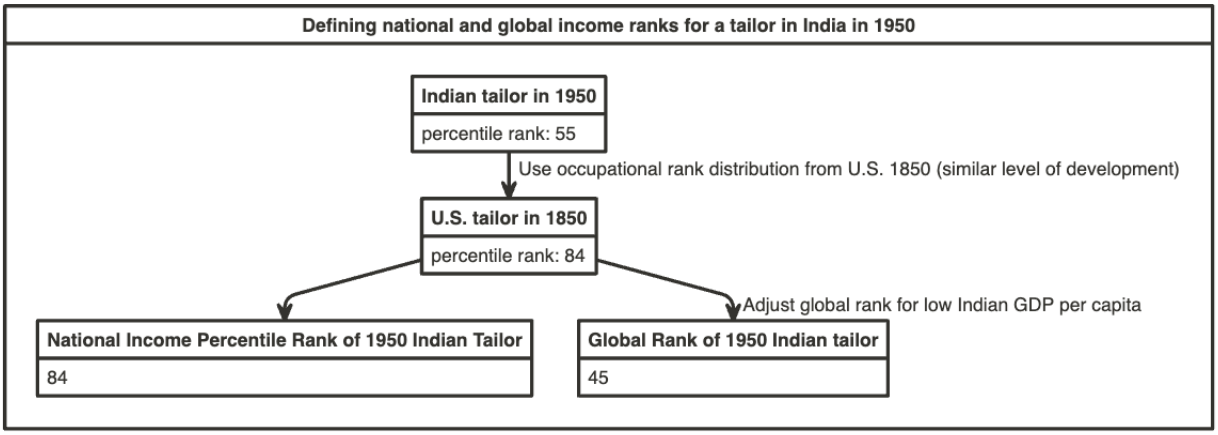


Table B1: List of occupational categories present in the analysis sample

Occupation Title	Frequency	Percent
Commercial Managers	105	15.09
Health professionals	81	11.64
Professors and instructors	69	9.91
Engineers	60	8.62
Natural scientists	34	4.89
Elementary and secondary school teachers	32	4.60
Jurists	28	4.02
Officials (government and non-profit organizations)	28	4.02
Farmers and farm managers	23	3.30
Sales workers and shop assistants	22	3.16
Accountants	20	2.87
Workers in religion	17	2.44
Bricklayers, carpenters, and related construction workers	14	2.01
Insurance agents	10	1.44
Other agents	9	1.29
Journalists, authors, and related writers	8	1.15
Protective service workers	8	1.15
Creative artists	7	1.01
Statistical and social scientists	7	1.01
Office and clerical workers	6	0.86
Operatives and kindred workers	6	0.86
Professional, technical, and related workers	6	0.86
Tailors and related workers	6	0.86
Welders and related metal workers	6	0.86
Architects	5	0.72
Forestry workers	5	0.72
Members of the armed forces	5	0.72
Bakers	4	0.57
Blacksmiths and machinists	4	0.57
Building managers and proprietors	4	0.57
Craftsmen and kindred workers	4	0.57
Jewelers, opticians, and precious metal workers	4	0.57
Locomotive operators	4	0.57
Other mechanics	3	0.43
Plumbers and pipe-fitters	3	0.43
Real estate agents	3	0.43
Ship officers	3	0.43
Social and welfare workers	3	0.43
Telephone operators	3	0.43
Textile workers	3	0.43
Vehicle mechanics	3	0.43
Aircraft pilots and navigators	2	0.29
Bookkeepers and related workers	2	0.29
Printers and related workers	2	0.29
Truck drivers	2	0.29
Butchers	1	0.14
Electronic service and repair workers	1	0.14
Farm laborers	1	0.14
Food service workers	1	0.14
Foremen	1	0.14
Gardeners	1	0.14
Launderers and dry-cleaners	1	0.14
Miners and related workers	1	0.14
Newsboys and deliverymen	1	0.14
Nonmedical technicians	1	0.14
Painters	1	0.14
Sawyers and lumber inspectors	1	0.14
Transport conductors	1	0.14

Notes: The table lists the occupational categories (according to the 1950 census classification) in our analysis sample and their frequency among the Nobel winners' fathers.

C Simulation: How much of the unequal distribution can be explained by heritability?

If scientific talent is heritable and parent talent contributes to higher parent income, then laureates will have high-SES parents even in a world with equal opportunity. An unequal outcome distribution is thus not proof of unequal opportunity. To what extent could genetic factors alone explain the unequal empirical laureate SES distribution?

We shed light on this question by simulating a world with equal opportunity but unequal genetic endowments, and we compare the parent SES distributions of top achievers to what we found in the data above. The simulation is based on the model in Section 4.1, but shuts down every channel through which parent income and child achievement are correlated, except the genetic channel.[30] We set $\beta_I = 0$, so that productive investments in children are not dependent on parent resources. We set $\delta = 0$, so unproductive investments do not contribute to career opportunity. Parent resources are thus correlated with scientific achievement only through the genetic inheritance of talent. This is an equal opportunity world, because career opportunities depend only on individual talent and some random shocks.

We add several features to our model to make it a better match for reality; these were left out of the model because they would not change the core conclusions and edge cases above. We simulate a population of parents with $N(0, 1)$ innate scientific talent (henceforth talent) and assortative mating.[31] Income-generating genes are positively but imperfectly correlated with talent and explain some share of father income, while the rest is explained by a random shock. We use a standard heritability model to simulate the scientific talent of children, given a heritability parameter h^2: children's talent is a function of parent talent and a random shock. The most talented children must pass through a set of filters to be eligible for prizes: a subset of them enter the basic sciences, a subset of those make breakthrough discoveries, and a subset of those are awarded Nobel prizes. We

[30] We could think about the simulation as representing a world where public education provides all the complementary inputs needed for talented children to maximally succeed.

[31] Assortative mating is important because it increases the correlation of ability between fathers and children.

calibrate the number of children and the number of prizes to the 1900–1950 United States birth cohorts, during which there were about 70 million boys born, of whom 196 would become laureates in the sciences.[32] Without the career entry and success filters, we would award prizes to the 196 most talented children in the sample; the more that are filtered out through career choice, random failure and prize error, the deeper into the talent distribution we will go to draw the winners.

The simulation generates an empirical SES distribution of Nobel laureate fathers, which we then compare to the empirical distribution calculated above. The simulation tells us, under a given set of parameters, how much of the observed inequality could be explained by genetic factors alone.

We estimate the simulation under three different parameter sets:

1. In the *mainline scenario*, we select parameters based on our reading of the literature that are intended to capture the midpoint of expert beliefs.

2. In the *nature-leaning* scenario, we set the parameters to values that lean toward finding that genes are important, at what we view as the outer bounds of parameter plausibility.

3. In the *nurture-leaning* scenario, we similarly set parameters toward their plausible limits in the direction of making genes unimportant determinants of scientific success.

The table below shows our chosen parameters under the three scenarios; our choices are explained below.

[32] With only 5 U.S.-born female laureates in this period, we ignore for now the obvious inequality by gender and aim to reproduce the SES inequality among men.

Table C1: Selected Simulation Parameters

Parameter	Nature-Leaning	Mainline	Nurture-Leaning
Heritability of scientific talent (h^2)	0.7	0.5	0.3
Correlation between realized income and genetic income potential	0.6	0.5	0.4
Genetic correlation between traits for income and scientific talent	0.9	0.8	0.7
Share of top talent entering basic sciences	10%	5%	2.5%
Failure rate among top scientific talent	50%	75%	85%
Nobel committee error in ranking scientists	25%	35%	45%
Assortative mating (father/mother talent correlation)	0.4	0.3	0.2

Heritability. Twin studies consistently estimate the heritability of cognitive skills (typically measured by a test or IQ score) around 0.5 (Bowles and Gintis, 2002; Polderman et al., 2015). Studies identifying specific genes for cognitive skills (*i.e.* genome-wide association studies, or GWAS, estimate much lower values in the range of 0.1–0.2 — this is the missing heritability problem (Howe et al., 2022; Savage et al., 2018). Estimates for the heritability of socioemotional skills and educational attainment are consistently lower in both methods, and neither method fully isolates gene-environment interactions (Beauchamp, 2016).[33] We set h^2 to 0.5 in the mainline scenario, 0.7 in the nature-leaning scenario, and 0.3 in the nurture-leaning scenario.

Correlation between father's genetic potential and actual income. The income generation process involves luck as well as talent, and a person with genetic potential that favors high income does not necessarily become high income. Bowles and Gintis (2002) find an elasticity of 0.15 between test scores and income; a meta-analysis by Strenze (2007) reports only a 0.21 correlation

[33]For example, girls with genes that cause them to go through puberty earlier have worse educational attainment. This loads on the heritability of educational attainment in all of these methods, but is probably caused by the social environment.

between IQ and adult income. These are underestimates of our parameter because IQ is not a perfect proxy of the set of genes that predict income. Twin studies find heritability of income between 0.3–0.5.[34] We set the parameter to 0.5 in the mainline scenario.

Correlation between income and scientific talent. Scientific success requires abilities and traits that differ from those that favor high earnings. There is substantial overlap in characteristics like intelligence and conscientiousness, but the optimal level of introversion is probably different in business and in science. We have wide priors on the genetic correlations between the relevant skillsets, so we set the mainline parameter conservatively to 0.8.

Selection into science. Fewer than 1% of adults work as scientists, professors, or engineers, and a very small fraction of these pursue non-incremental basic science where a Nobel Prize would even be possible. Many talented individuals choose more lucrative careers in medicine, law, or business. Our mainline assumption is that 5% of the most talented children join careers in the basic sciences.

Failure rate. Even exceptionally talented scientists may not achieve breakthrough discoveries due to research direction choices, or simply bad luck either in research or in life.[35] Since failure in this case means anything short of a Nobel-worthy discovery, we set the mainline failure rate to 75%.

Nobel committee error. The Nobel Prize selection process involves judgment and is subject to error in identifying the most talented scientists. There are multiple famous cases of overlooked scientists whose contributions were later recognized as groundbreaking. The mainline scenario assigns the committee a 35% error rate.

[34] Hyytinen et al. (2019) finds income heritability of 0.5 in Finnish twins. Horwitz et al. (2003) finds evidence that twin study heritability estimates are biased upwards, as monozygotic and dizygotic twins do not experience the same degree of shared environment. Bingley et al. (2024) finds heritability of only 0.15 among Danish twins after accounting for differences in environment between monozygotic and dizygotic twins.

[35] Bad luck could mean something destabilizing like a serious negative health shock, or could amount to simply choosing the wrong field, like being a brilliant organic chemist in the molecular biology era, or being outstanding at optics in the quantum mechanics era.

Assortative mating. If people marry partners with similar genes, their children will have more genetic similarity to themselves. Empirical studies find high correlations of education between spouses, but much lower correlations for specific skills. A large study in Britain (Horwitz et al., 2023) found a correlation of 0.55 in educational attainment between partners, but 0.44 in IQ and only 0.16 in conscientiousness, all important characteristics for success in the sciences. A recent study from Norway finds similar estimates, and argues for lower genotype than phenotype correlations (Sunde et al., 2025). We set the spousal correlation of latent scientific talent to 0.3 in the mainline scenario.

To provide an external check on our parameterizations, we gave detailed prompts to the four major AI reasoning models and asked them to select mainline, nature-leaning and nurture-leaning parameters. The models' parameter choices are summarized in Appendix Table C2, along with links to the full prompts and results. The AI model choices were similar to ours overall; the chief difference is that they chose much higher failure rates. Section C.2 describes the complete structure and execution of the simulation, and provides a link to the simulation code.

Table C2: AI-generated Simulation Parameters

Parameter	OpenAI o3	Gemini 2.5 Pro	Claude Sonnet 4.0	Grok 3.0
Heritability of scientific talent (h^2)	0.55	0.65	0.60	0.70
Genetic correlation between traits for income and scientific talent	0.55	0.60	0.65	0.67
Correlation between father's income and child's genetic income	0.45	0.50	0.45	0.45
Share of top talent entering basic sciences	15%	20%	15%	50%
Failure rate among top scientific talent	97%	90%	85%	90%
Nobel committee error in ranking scientists	35%	33%	25%	10%
Assortative mating (father/mother talent correlation)	0.3	0.50	0.40	0.40

Notes: We gave detailed prompts to the four leading reasoning AI models. Reasoning models go beyond standard language models by conducting web and literature searches and conducting iterative evaluations of results. Our prompts and the model responses can be found at the following URLs: OpenAI o3, Claude Sonnet 4.0, Gemini Pro 2.5, Grok 3.0. If these links do not work, they can be found at `https://github.com/devdatalab/nobel-sim`. Repeated draws to the same models generate variation in simulation parameters but they do not differ wildly from the results shown here. Parameterizing models using suggestions from AI models is a speculative exercise, but we expect it to represent some aggregation of scientific consensus and widely-held beliefs about the parameters in question.

C.1 Simulation Results

We summarize the simulation results in Table C3, focusing on the simulated share of laureates with fathers in the bottom 80% and 90% of the income distribution, respectively. Figure C1 shows sample distribution plots from our three scenarios, as well as the comparison empirical income distribution of laureate fathers.

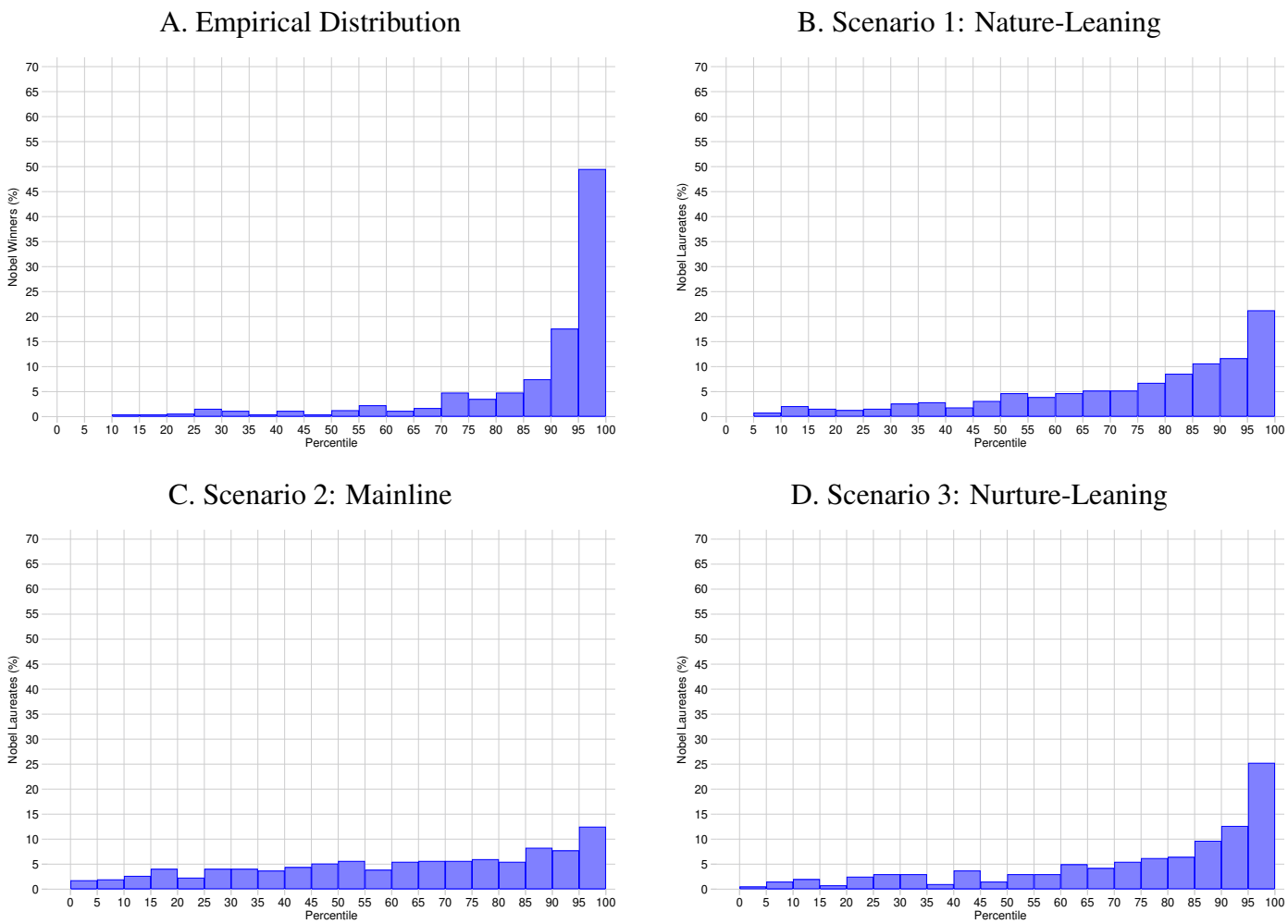
In the empirical distribution, 21% of laureate fathers are from bottom 80% households. In our mainline simulation, 64% are from bottom 80% households. In other words, strictly genetic factors can explain 26% of the empirical inequality in the SES distribution of laureate fathers (*i.e.* (80–64)/(80–21)). Similarly, we find strictly genetic factors can explain only 19% of the inequality in the share of fathers from the bottom 90%. These estimates are similar to Bell et al. (2019) who argue that raw ability (proxied by 3rd and 8th grade test scores) explains less than a third of the rich-poor gap in patent production. When we tip all the parameters toward finding a role for

genetics, then genes can explain 54–66% of the inequality, while genes in the nurture-leaning model explain only 6–8% of the inequality. The AI models' mainline parameterizations also supported our mainline result that genes can explain between a third and a fifth of the inequality.[36] Finally, we note that genes can explain an even smaller share of the global inequality in scientific access, given that only 5% of laureates are from the bottom 80% globally.

The model in Section 4.1 showed that a high correlation between parent income and child scientific success was not in itself proof of lost scientific output, because high-income parents may pass high-ability genes and productive educational investments onto their children. This section demonstrates how different primitive assumptions translate into different distributions of SES backgrounds among Nobel laureates. We continue to have wide priors on some simulation parameters and the results show that a range of beliefs about the role of genes in the inequality of scientific achievement is consistent with the data. More research aiming to estimate the underlying simulation parameters could improve the precision of this exercise.

[36]The AI model results should not be treated as arbiters of truth, as these models can hallucinate and contain unknown biases. Rather, we show these results as evidence that our parameterizations are reasonably in line with an aggregation of expert and popular belief. The models' parameterizations (see Appendix Table C2) were very close to ours, choosing slightly higher heritability values, but much higher failure rates.

Figure C1: Simulating the distribution of laureate SES ranks under equal opportunity assumptions



Notes: Panel A shows the empirical distribution of the income ranks of fathers of the Nobel laureates, reproducing Figure 1A. The remaining figures are simulation results from a model where father income and child scientific output are linked exclusively through the genetic inheritance of talent; all unequal investment channels are shut down. Panel B shows the result under parameters set to maximal reasonable values to emphasize the role of genes. Panel C shows our mainline estimates. Panel D shows the results with parameters set to emphasize the role of parent investments. Each graph shows a result from a single run of the simulation.

Table C3: Simulated shares of Nobel laureates with fathers below the 80th and 90th income percentiles

	Share of parents in bottom 80%	Share of parents in bottom 90%
Nature-leaning	36.7	53.2
	(2.9)	(1.9)
Mainline	62.7	76.9
	(1.9)	(1.8)
Nurture-leaning	72.8	84.8
	(1.6)	(1.1)
Empirical	20.7	32.9

Notes: The table reports estimates from a simulation of father income and child entry and success in the sciences. The estimates show the share of laureates whose fathers fall below the 80th and 90th income percentiles under each set of parameters.

C.2 Implementation of the Simulation Model

Our simulation models a counterfactual world in which individuals succeed purely on the basis of heritable cognitive ability. The simulation is based on the model in Section 4.1, but we use a different normalization for the genetic and environmental components of inheritance, basing our terminology here on standards in the genetics literature.

Children's ability is partially inherited from parents and partially the result of random shocks. Nobel prizes are awarded to the highest ability children who (i) enter the basic sciences; (ii) make breakthrough discoveries; and (iii) are recognized by the prize committee.

The simulation represents 70 million male births in U.S. cohorts 1900–1950, of which 196 received Nobel prizes.

Scientific ability phenotypes. Each individual has a latent scientific ability phenotype (P), part of which arises from additive genetic effects (G), where heritability is h^2. E is the non-inherited component and is orthogonal to G.

$$P = G + E, \qquad \text{where} \quad G \sim \mathcal{N}(0, h^2), \qquad E \sim \mathcal{N}(0,\, 1 - h^2), \qquad G \perp E. \tag{C1}$$

Parents. For every birth, we draw a mother and father, with assortative correlation a:

$$\begin{aligned} P_m, P_f &\sim \mathcal{N}(0, 1), \\ \text{corr}(P_m, P_f) &= a. \end{aligned} \tag{C2}$$

We construct the correlation a on the phenotype by drawing:

$$\begin{aligned} G_f &\sim \mathcal{N}(0, h^2), & E_f &\sim \mathcal{N}(0, 1 - h^2); \\ G_m &= a\, G_f + \sqrt{1 - a^2}\, \varepsilon_G, & \varepsilon_G &\sim \mathcal{N}(0, h^2); \\ E_m &= a\, E_f + \sqrt{1 - a^2}\, \varepsilon_E, & \varepsilon_E &\sim \mathcal{N}(0, 1 - h^2); \\ P_f &= G_f + E_f, \quad P_m = G_m + E_m. \end{aligned} \tag{C3}$$

with independent $\varepsilon_G, \varepsilon_E \sim \mathcal{N}(0, h^2)$ and $\mathcal{N}(0, 1 - h^2)$ respectively.

Father income. Fathers have a genetic phenotype I_f, which is a predictor of high income and has correlation γ with latent scientific ability. The father's standardized income Y_f is linked to this phenotype with correlation ρ, a model parameter which captures the fact that random shocks make up a substantial part of a person's income realization:

$$I_f \;=\; \gamma\, P_f + \sqrt{1 - \gamma^2}\, \xi, \qquad \xi \sim \mathcal{N}(0, 1). \tag{C4}$$

$$Y_f \;=\; \rho\, I_f + \sqrt{1 - \rho^2}\, \eta, \quad \eta \sim \mathcal{N}(0, 1). \tag{C5}$$

Child scientific ability. A child's latent scientific ability P_c has an additive genetic component G_c derived from the parent and an environmental component E_c. Because this is an equal-opportunity world, E_c is independent of parental traits:

$$G_c \;=\; \tfrac{1}{2}\big(G_m + G_f\big) \;+\; \varepsilon_M, \quad \varepsilon_M \sim \mathcal{N}\Big(0, \tfrac{1}{2}(1-a)h^2\Big).$$

(C6)

$$E_c \sim \mathcal{N}(0,\, 1-h^2), \quad E_c \perp \{G_c, P_m, P_f, Y_f\}.$$

(C7)

$$P_c = G_c + E_c \sim \mathcal{N}(0, 1).$$

(C8)

Note that we use additive (narrow-sense) heritability, because only additive variance results in deterministic parent-offspring covariance. Many headline "heritability" numbers in the behavior-genetics literature are broader and include epistatic and dominance effects which are not reliably transmitted to children. As such, a twin study estimate of heritability mixing additive and non-additive components would need to be adjusted downward to be used in this model.

Mapping Ability to Nobel Status. The child must clear three independent filters:

1. **Entry:** Chooses a science career. Retention probability: e
2. **Survival:** Avoids failure or dropout. Retention probability: s

3. **Committee:** Not mis-ranked by Nobel jury. Retention probability: r

Combined retention $\pi = e \cdot s \cdot r$. Let Φ be the standard normal CDF. Define the cutoff $z_\star$ by

$$\Pr\big(P_c > z_\star\big) \;=\; \frac{L}{N\pi} \;\Longrightarrow\; z_\star = \Phi^{-1}\Big(1 - \frac{L}{N\pi}\Big).$$

(C9)

A child with ability $P_c > z_\star$ that survives the pipeline is labeled a laureate.

Recorded Statistics. For every run of the simulation, we store the following statistics for the set of children identified as laureates:

$$\mathbf{1}\{\mathrm{rank}(Y_f) < 80\%\}, \qquad \mathbf{1}\{\mathrm{rank}(Y_f) < 90\%\}.$$

(C10)

The procedure is repeated 10 times for each set of parameters, and we report the mean and standard deviation from the set of 10 runs in Table C3. This was a sufficient number of simulations to produce tight estimates that distinguished the three scenarios.

The complete code for the simulation can be found at `https://github.com/devdatalab/nobel-sim`.